# CSE4088 Introduction to Machine Learning

Introduction and Overview

References:
Lecture notes of Tom Mitchell, Aykut Erdem and Maria-Florina Balcan

# Quotes

- *"If you were a current computer science student what area would you start studying heavily?*
  - *–Answer: Machine Learning.*
  - *–"The ultimate is computers that learn"*

  –Bill Gates, Reddit AMA

- *"Machine learning is the next Internet"*

  –Tony Tether, Director, DARPA

- *"Machine learning is today's discontinuity"*

  –Jerry Yang, CEO, Yahoo

- *"AI is the new electricity! Electricity transformed countless industries; AI will now do the same."*

  –Andrew Ng

slide by David Sontag

# Machine Learning is Changing the World

"Machine learning is the hot new thing"
(John Hennessy, President, Stanford)





"A breakthrough in machine learning would be worth ten Microsofts" (Bill Gates, Microsoft)

"Web rankings today are mostly a matter of machine learning"
(Prabhakar Raghavan, VP Engineering at Google)









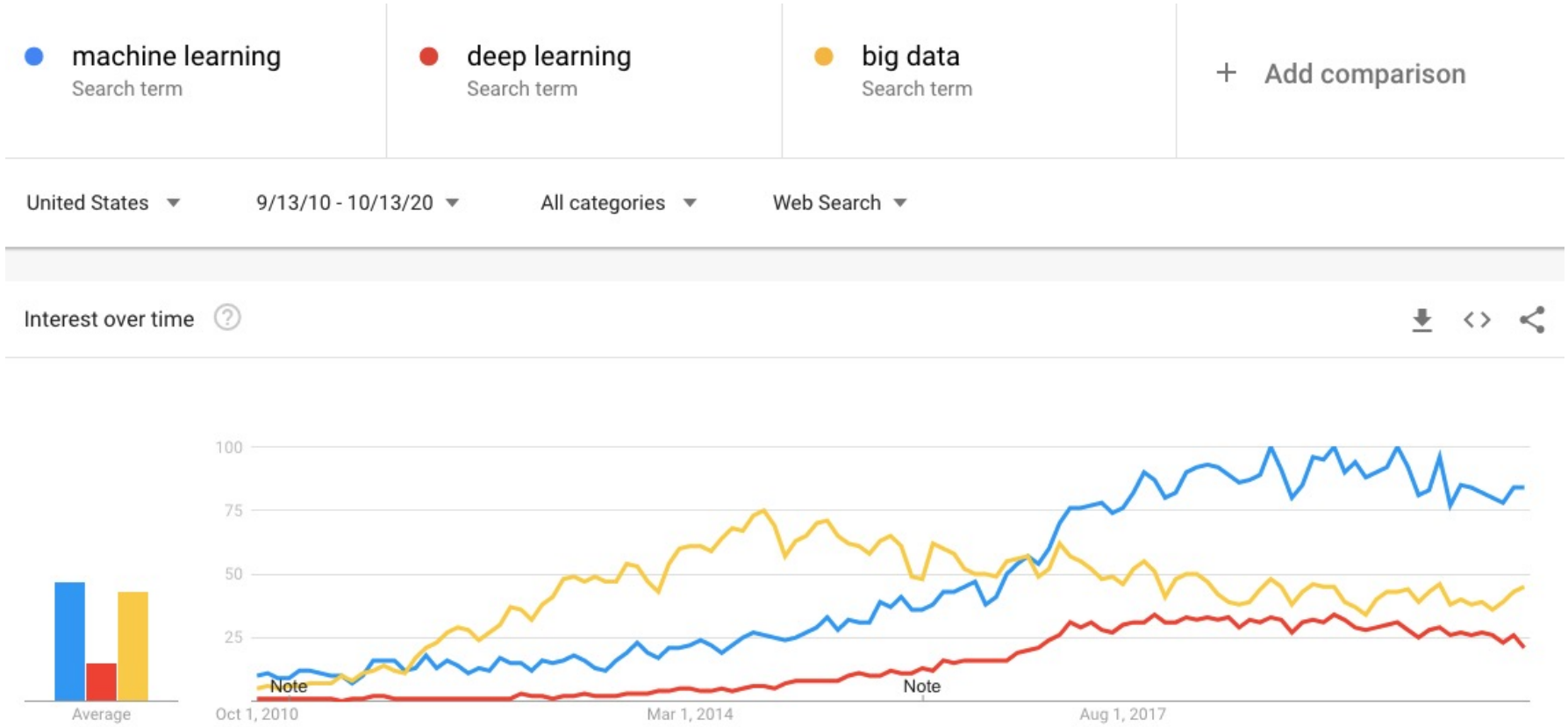


# The COOLEST TOPIC IN SCIENCE

- "A breakthrough in machine learning would be worth ten Microsofts" (Bill Gates, Chairman, Microsoft)
- "Machine learning is the next Internet" (Tony Tether, Director, DARPA)
- Machine learning is the hot new thing" (John Hennessy, President, Stanford)
- "Web rankings today are mostly a matter of machine learning" (Prabhakar Raghavan, Dir. Research, Yahoo)
- "Machine learning is going to result in a real revolution" (Greg Papadopoulos, CTO, Sun)
- "Machine learning is today's discontinuity" (Jerry Yang, CEO, Yahoo)

# Google Trends

(trends.google.com)



# Machine Intelligence LANDSCAPE 2015 Edition

## CORE TECHNOLOGIES

### ARTIFICIAL INTELLIGENCE
IBM WATSON, MetaMind, Numenta, ai-one, Cycorp, Microsoft Research, nara, Reactor, SI SCALED INFERENCE

### DEEP LEARNING
vicarious, Vision Factory, facebook, Baidu IDL, Google, ersatz labs, SKYMIND, SignalSense

### MACHINE LEARNING
rapidminer, context relevant, 0xdata H2O, DATARPM, LIFTIGNITER, SPARKBEYOND, Azure ML, yhat, wise.io, Sense, GraphLab, Alpine, tonian

### NLP PLATFORMS
cortical.io, idibon, LUMINOSO, wit.ai, Maluuba

### PREDICTIVE APIS
AlchemyAPI, MINDOPS, Google, bigml, indico, ALGORITHMIA, Expect Labs, PredictionIO

### IMAGE RECOGNITION
clarifai, MADBITS, DNNresearch, DEXTRO, VISENZE, lookflow

### SPEECH RECOGNITION
GRIDSPACE, popup archive, NUANCE

## RETHINKING ENTERPRISE

### SALES
Preact, AVISO, RelateIQ, NGDATA, CLARABRIDGE, FRAMED, infer, ATTENSITY, causata

### SECURITY / AUTHENTICATION
CROSSMATCH, conjur, EYEVERIFY, BITSIGHT, CYLANCE, AREA 1 SECURITY, bionym

### FRAUD DETECTION
sift science, SOCURE, ThreatMetrix, feedzai, Brighterion, VERAFIN

### HR / RECRUITING
TalentBin, entelo, predikt, Connectifier, gild, hiQ, CONCEPTNODE

### MARKETING
brightfunnel, bloomreach, CommandIQ, AIRPR, RADIUS, TellApart, people pattern, Freshplum

### PERSONAL ASSISTANT
Siri, Google now, Cortana, cleversense, tempo, Robinlabs, KASISTO, fusemachines, VIV, CLARA LABS

### INTELLIGENCE TOOLS
ADATAO, Palantir, Quid, Digital Reasoning, FirstRain

## RETHINKING INDUSTRIES

### ADTECH
METAMARKETS, dstillery, rocketfuel, YieldMo, ADBRAIN

### AGRICULTURE
BLUE RIVER, TerrAvion, ceresimaging, HONEYCOMB, THE CLIMATE CORPORATION, tule, mavrx

### EDUCATION
declara, coursera, KNEWTON, kidaptive

### FINANCE
Bloomberg, FinGenius, alphasense, KENSHO, Dataminr, minettabrook, BINATIX

### LEGAL
Lex Machina, brightleaf, COUNSELYTICS, RAVEL, JUDICATA, Brevia, DiligenceEngine

### MANUFACTURING
SIGHT MACHINE, MICROSCAN, BOULDER IMAGING, IVISYS

### MEDICAL
Parzival, transcriptic, ZEPHYR, Genescient, grand round table, bina, TUTE GENOMICS

### OIL AND GAS
kaggle, AYASDI, TACHYUS, biota, Flutura

### MEDIA / CONTENT
Outbrain, newsle, ARRIA, SAILTHRU, wavii, Owlin, NarrativeScience, yseop, Summly, Prismatic, ai

### CONSUMER FINANCE
affirm, inVenture, zestfinance, BILLGUARD, LendUp, LendingClub, Kabbage

### PHILANTHROPIES
DataKind, thorn, THE DATA GUILD

### AUTOMOTIVE
Google, Continental, TESLA, MOBILEYE, CRUISE

### DIAGNOSTICS
enlitic, 3SCAN, lumiata, ENTOPSIS

### RETAIL
BAY SENSORS, PRISM SKYLABS, celect, euclid

## RETHINKING HUMANS / HCI

### AUGMENTED REALITY
wearable intelligence, APX, blippar, META, layar

### GESTURAL COMPUTING
THALMICLABS, omek, FLUTTER, LEAP MOTION, eyeSight, 3Gear systems, GestureTek, nod

### ROBOTICS
intel, LIQUID ROBOTICS, SoftBank, iRobot, Boston Dynamics, jibo, anki, evolution robotics

### EMOTIONAL RECOGNITION
affectiva, BEYONDVERBAL, EMOTIENT, cogito

## SUPPORTING TECHNOLOGIES

### HARDWARE
NVIDIA, XILINX, QUALCOMM, NERVANA SYSTEMS, TERADEEP, Artificial Learning, rigetti

### DATA PREP
TRIFACTA, Paxata, tamr, Alation

### DATA COLLECTION
diffbot, kimono, CrowdFlower, Connotate, WorkFusion, import.io

www.shivonzilis.com/machineintelligence

Bloomberg BETA

# MACHINE INTELLIGENCE 2.0

# 2016 Edition

## AGENTS

### PROFESSIONAL

Howdy! x.ai clara
KASIST@ DigitalGenius
OVERLAP.CC meekan
fuse|machines PRIMER

### PERSONAL

facebook XIAOICE
large
assistant.ai nestor
@wesome Magic

### OS INTERFACES

Siri Cortana VIV
maluuba api.ai
CogNEA Google Now

## AUTONOMOUS SYSTEMS

### AIR

3DR DJI PROJECT LOON
VERTICAL DroneDeploy
AIRDOG SKYCATCH
SKYDIO Airware LILY

### GROUND

Google UBER
TESLA CRUISE
MOBILEYE
COMMA AdasWorks

### SEA

LIQUID ROBOTICS
bluefin data
OPENROV
BluHaptics

### INDUSTRIAL

KIVA Systems fetch
HARVEST CLEARPATH
AVIDBOTS ENERGID
rethink robotics GREYORANGE
OSARO

## ENTERPRISE

### SECURITY / FRAUD

Sentinel graphistry
BITSIGHT feedzai AREA1
drawbridge sift science
CYLANCE Brighterion

### HR / RECRUITING

textio hiQ gild
SpringRole entelo
unitive GIGSTER

### SALES

6sense infer people pattern
Preact Prism AVISO
Vidora sentient
salespredict Gainsight

### MARKETING

LiftIgniter
RADIUS brightfunnel
retention SCIENCE AIRPR

### CUSTOMER SUPPORT

CLARABRIDGE
QUANTIFIND Wisejo
ACTIONIQ FRAMED
DigitalGenius

### INTERNAL INTEL

Alation ADATAO
Palantir sapho lucid
Rainbird SKIPFLAG Cogito
Digital Reasoning Narrative Science

### MARKET INTEL

Quid mattermark
DataFox bottlenose
PREMISE enigma
CB INSIGHTS

## PLATFORMS

### RESEARCH / AGI

OpenAI vicarious
Google DeepMind Numenta
Cycorp nnaisense
SCALED INFERENCE CURIOUS
GEOMETRIC INTELLIGENCE

### FULL STACK

context relevant
CognitiveScale
NVIDIA TERADEEP
QUALCOMM nervana SYSTEMS

### MACHINE LEARNING

Dato rapidminer
cortical.io AYASDI
amazon web services Azure Machine Learning
narrologics PredictionIO
SKYTREE bigml blueyonder

### INDUSTRIAL IOT

ThingWorx UPTAKE
IMUBIT Preferred Networks
Alluvium xively
PLANET OS

### AUDIO

Gridspace TalkIQ
nexidia vocaliq
NUANCE Expect Labs
popup archive

### VISION

ORBITAL INSIGHT
Descartes Labs DEXTRO
cortica clarifai PLANET LABS
MetaMind

### DATA ENRICHMENT

diffbot Paxata
TRIFACTA iDIBON
WorkFusion loop AI LABS
CrowdFlower

## INDUSTRIES

### ADTECH

ADTHEORENT dstillery
BEYONDVERBAL
METAMARKETS TAPAD
rocketfuel affectiva

### AGRICULTURE

BLUE RIVER tule
TerrAvion mavrx
THE CLIMATE CORPORATION CERES TECHNOLOGIES
HONEYCOMB

### FOR GOOD

Conservation Metrics
DataKind DATA POP
thorn
BAYES IMPACT

### RETAIL FINANCE

inVenture affirm
earnest MIRADOR
Lendo zest finance LendUp

### LEGAL

Everlaw RAVEL
LEGAL ROBOT Seal
BEAGLE ROSS
Lex Machina

### MATERIALS & MFG

zymergen AUGMATE
GINKGO BIOWORKS
ITRINE TECHNOLOGIES SIGHT MACHINE
CALCULARIO
Eigen Innovations

### HEALTHCARE

deep genomics 3SCAN
enlitic Calico BUTTERFLY NETWORK, INC.
Atomwise Recombine color
METABIOTA GRAND ROUNDS
Google Life Sciences IBM Watson Health

## INDUSTRIES (CONT'D)

### EDUCATION

KNEWTON
coursera turnitin
gradescope UDACITY
KHANACADEMY

### TRANSPORT & LOGISTICS

NAUTO taleris
PRETECKT
CM clearmetal

### INVESTMENT FINANCE

Bloomberg
Quantopian
Dataminr KENSHO
iSENTIUM NEURENSIC
alphasense

## TECH USER TOOLS

### DATA SCIENCE

DOMINO kaggle
Sentenai sense
yseop Outlier
ŷhat DataRobot

### MACHINE LEARNING

Cortana Analytics AlchemyAPI glowfi.sh
IBM Watson Platform Anodot MonkeyLearn
(h[s]) HyperScience fuzzy.io SIGOPT
0xdata H2O SPARKBEYOND indico

### OPEN SOURCE

SKYMIND TensorFlow
seldon Caffe theano
Spark MLlib Microsoft DMTK spaCy
DL4J SciKit CGT

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clarifai
DEEPVISION
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Captricity
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AUDIO
Gridspace
TalkIQ
nexidia
twilio
CAPIO
Expect Labs
Clover
Mobvoi
Qurious.AI
popUP archive
SENSOR
PREDIX
GE IoT
MAANA
Sentenai
PLANET OS
UPTAKE
IMUBIT
Preferred Networks
thingworx
KONUX
Alluvium
INTERNAL DATA
PRIMER
IBM WATSON
Cycorp
Palantir
ARIMO
Alation
sapho
Outlier
Digital Reasoning
MARKET
mattermark
Quid
DataFox
PREMISE
Bottlenose
MOTIVA
enigma
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Tracxn
predata
ENTERPRISE FUNCTIONS
CUSTOMER SUPPORT
DigitalGenius
Kasisto
ELOQUENT
Wise.io
ACTIONIQ
zendesk
Preact
CLARABRIDGE
SALES
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6sense
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AVISO
salesforce
INSIDE SALES .COM
clari
Zensight
MARKETING
MINTIGO
Lattice
RADIUS
Liftigniter
PERSADO
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SHIELD AI
Airware
dji
LILY
DroneDeploy
pilotAI
SKYCATCH
INDUSTRIAL
JAYBRIDGE
OSARO
CLEARPATH
fetch ROBOTICS
KINDRED
HARVEST AUTOMATIC
rethink robotics
AGENTS
PERSONAL
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Cortana
Allo
facebook
Siri
Replika
PROFESSIONAL
butter.ai
pogo
SKIPFLAG
clara
x.ai
slack
talla
Zoom.ai
sudo
INDUSTRIES
AGRICULTURE
BLUE RIVER
mavrx
tule
TRACE GENOMICS
Pivot Bio
TerrAvion
AGRI-DATA
Descartes Labs
udelv
abundant ROBOTICS
EDUCATION
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volley
gradescope
CTI
coursera
UDACITY
altschool
INVESTMENT
Bloomberg
sentient
iSENTIUM
KENSHO
alphasense
Dataminr
CEREBELLUM CAPITAL
Quandl
LEGAL
blueJ
BEAGLE
Everlaw
RAVEL
Seal
ROSS
LEGAL ROBOT
LOGISTICS
NAUTO
Acerta
PRETECKT
clearmetal
Routific
MARBLE
PITSTOP
INDUSTRIES CONT'D
MATERIALS
zymergen
Citrine
Eigen Innovations
SIGHT MACHINE
GINKGO BIOWORKS
nanotronics
CALCULARIO
RETAIL FINANCE
TALA
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Lendo
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Betterment
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Oncora
SENTRIAN
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IMAGE
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3SCAN
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BAYLABS
imagia
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deep genomics
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kaggle
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AYASDI
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seldon
yseop
bigml
MACHINE LEARNING
CognitiveScale
GoogleML
context relevant
Cycorp
HyperScience
narrativelogics
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H2O.ai
SCALED INFERENCE
sparkcognition
Loop
GEOMETRIC INTELLIGENCE
deepsense.io
reactive
skymind
bonsai
NATURAL LANGUAGE
agolo
AYLIEN
LEXALYTICS
Narrative Science
spaCy
LUMINOSO
cortical.io
MonkeyLearn
DEVELOPMENT
SIGOPT
HyperOpt
fuzzyio
pkite
rainforest
lobe
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LAYER 6
bonsai
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CrowdAI
import.io
Paxata
DATASIFT
amazon mechanical turk
enigma
WorkFusion
DATALOGUE
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parsehub
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Keras
Chainer
CNTK
TensorFlow
Caffe
H20
DEEPLEARNING4J
theano
torch
DSSTNE
Scikit-learn
AzureML
neon
MXNet
DMTK
Spark
PaddlePaddle
WEKA
HARDWARE
KNUPATH
TENSTORRENT
Cirrascale
NVIDIA
nervana
Movidius
tensilica
GoogleTPU
10^26 Labs
Qualcomm
Cerebras
Isosemi
RESEARCH
OpenAI
maisense
ELEMENT AI
vicarious
KNOGGIN
Numenta
Kimera Systems
Cogitai

# What is Machine Learning?

# Example: Netflix Challenge

- Goal: Predict how a viewer will rate a movie
- 10% improvement = 1 million dollars

Year 2009







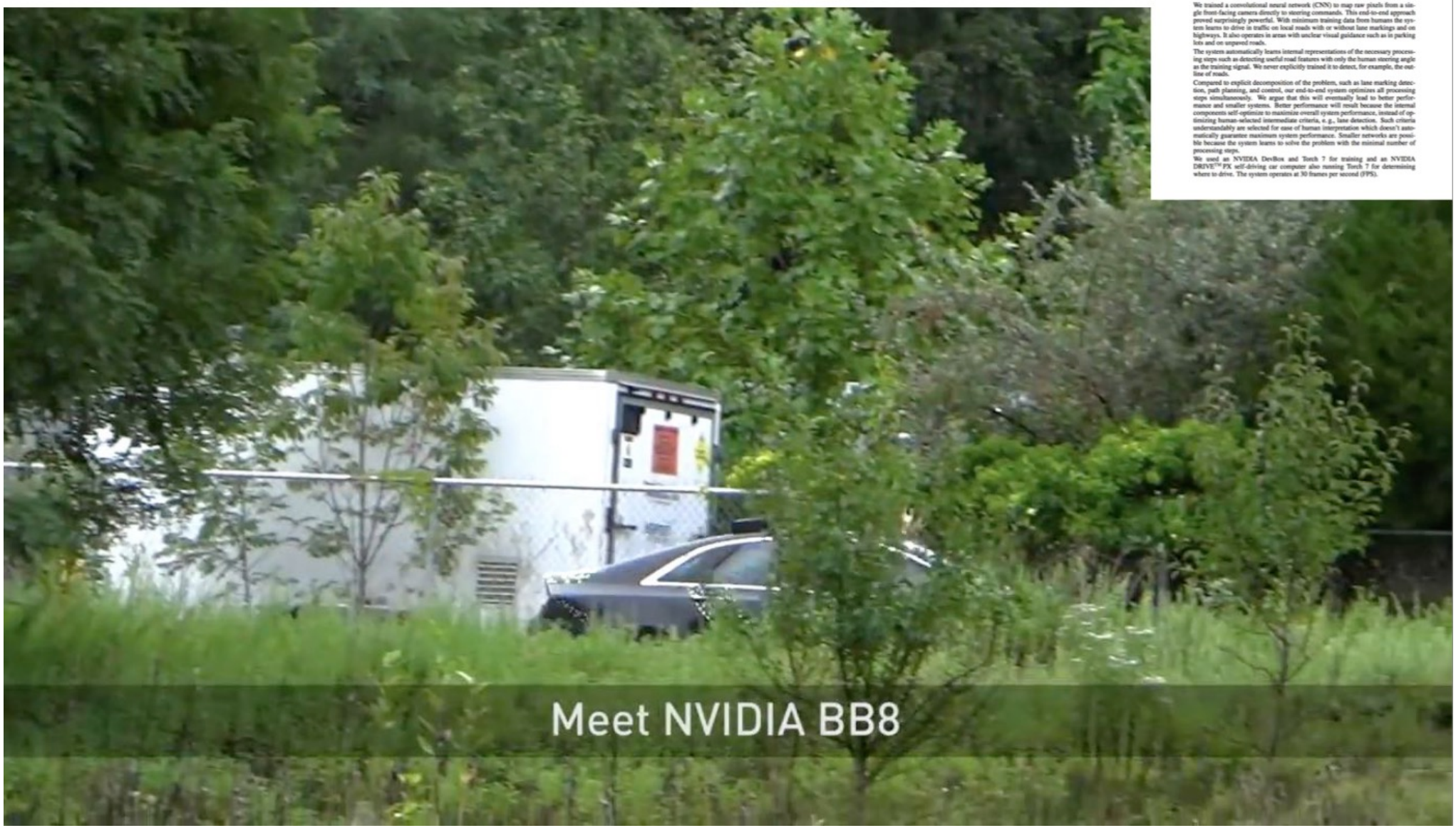
slide by Yaser Abu-Mostapha

# Example: Netflix Challenge

- Goal: Predict how a viewer will rate a movie
- 10% improvement = 1 million dollars
- Essence of Machine Learning:
  - A pattern exists
  - We cannot pin it down mathematically
  - We have data on it

slide by Yaser Abu-Mostapha

# NVIDIA BB8 AI Car




**End to End Learning for Self-Driving Cars**

**Mariusz Bojarski** NVIDIA Corporation Holmdel, NJ 07735
**Davide Del Testa** NVIDIA Corporation Holmdel, NJ 07735
**Daniel Dworakowski** NVIDIA Corporation Holmdel, NJ 07735
**Bernhard Firner** NVIDIA Corporation Holmdel, NJ 07735
**Beat Flepp** NVIDIA Corporation Holmdel, NJ 07735
**Prasoon Goyal** NVIDIA Corporation Holmdel, NJ 07735
**Lawrence D. Jackel** NVIDIA Corporation Holmdel, NJ 07735
**Mathew Monfort** NVIDIA Corporation Holmdel, NJ 07735
**Urs Muller** NVIDIA Corporation Holmdel, NJ 07735
**Jiakai Zhang** NVIDIA Corporation Holmdel, NJ 07735
**Xin Zhang** NVIDIA Corporation Holmdel, NJ 07735
**Jake Zhao** NVIDIA Corporation Holmdel, NJ 07735
**Karol Zieba** NVIDIA Corporation Holmdel, NJ 07735

**Abstract**

We trained a convolutional neural network (CNN) to map raw pixels from a single front-facing camera directly to steering commands. This end-to-end approach proved surprisingly powerful. With minimum training data from humans the system learns to drive in traffic on local roads with or without lane markings and on highways. It also operates in areas with unclear visual guidance such as in parking lots and on unpaved roads.

The system automatically learns internal representations of the necessary processing steps such as detecting useful road features with only the human steering angle as the training signal. We never explicitly trained it to detect, for example, the outline of roads.

Compared to explicit decomposition of the problem, such as lane marking detection, path planning, and control, our end-to-end system optimizes all processing steps simultaneously. We argue that this will eventually lead to better performance and smaller systems. Better performance will result because the internal components self-optimize to maximize overall system performance, instead of optimizing human-selected intermediate criteria, e. g., lane detection. Such criteria understandably are selected for ease of human interpretation which doesn't automatically guarantee maximum system performance. Smaller networks are possible because the system learns to solve the problem with the minimal number of processing steps.

We used an NVIDIA DevBox and Torch 7 for training and an NVIDIA DRIVE™ PX self-driving car computer also running Torch 7 for determining where to drive. The system operates at 30 frames per second (FPS).

# What is Machine Learning?

- [Arthur Samuel, 1959]
  - Field of study that gives computers
  - the ability to learn without being explicitly programmed

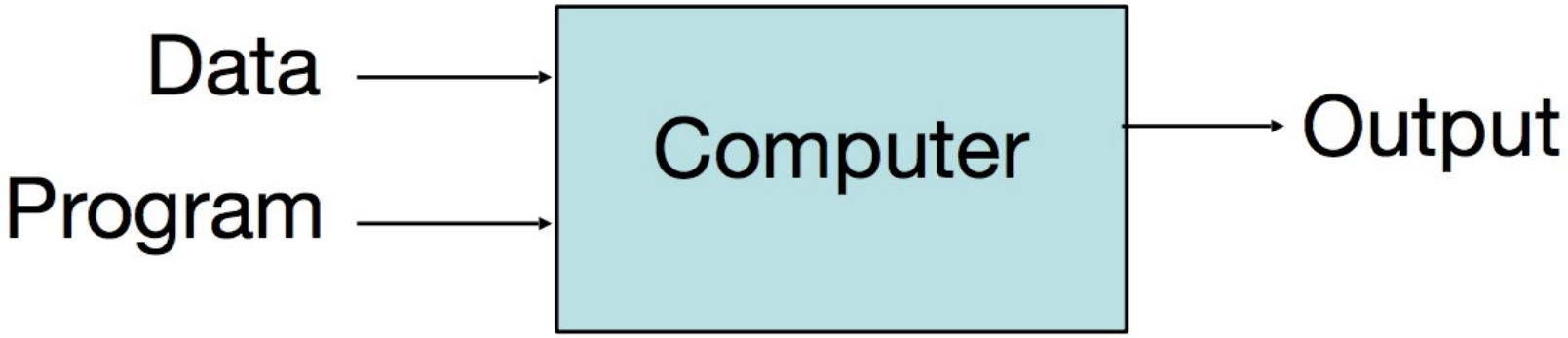
- [Kevin Murphy] algorithms that
  - automatically detect patterns in data
  - use the uncovered patterns to predict future data or other outcomes of interest

- [Tom Mitchell] algorithms that
  - improve their performance (P)
  - at some task (T)
  - with experience (E)

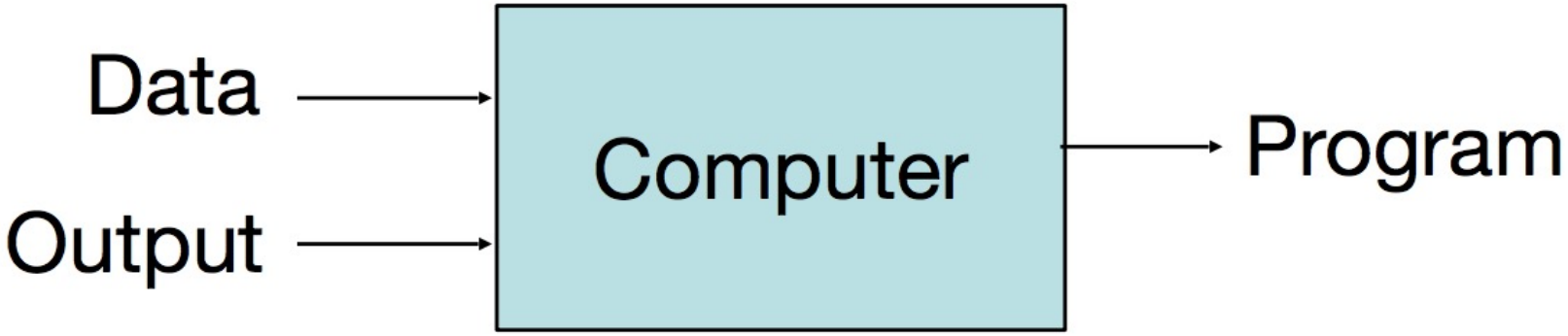
slide by Dhruv Batra

# Comparison

- **Traditional Programming**



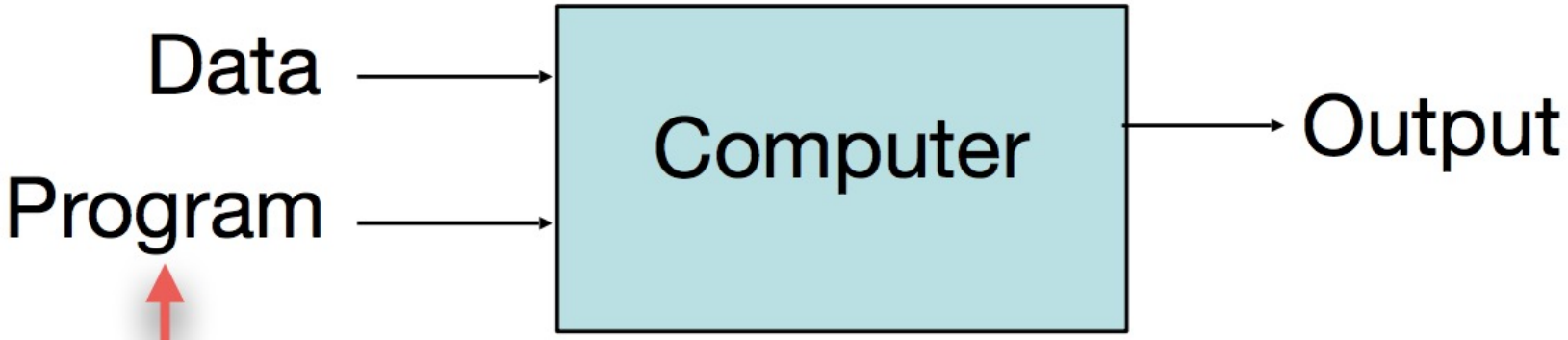

- **Machine Learning**



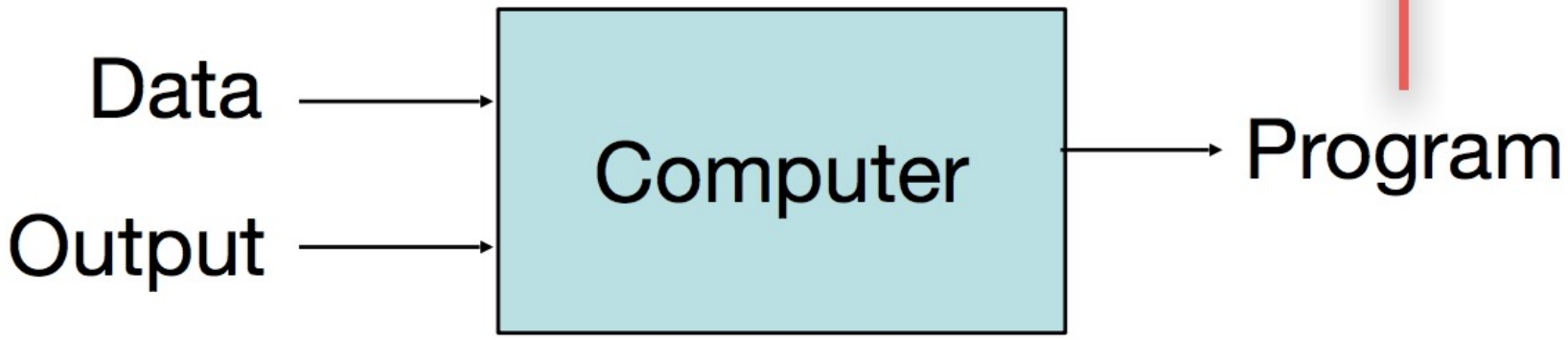

slide by Pedro Domingos, Tom Mitchel, Tom Dietterich

# Comparison

- **Traditional Programming**



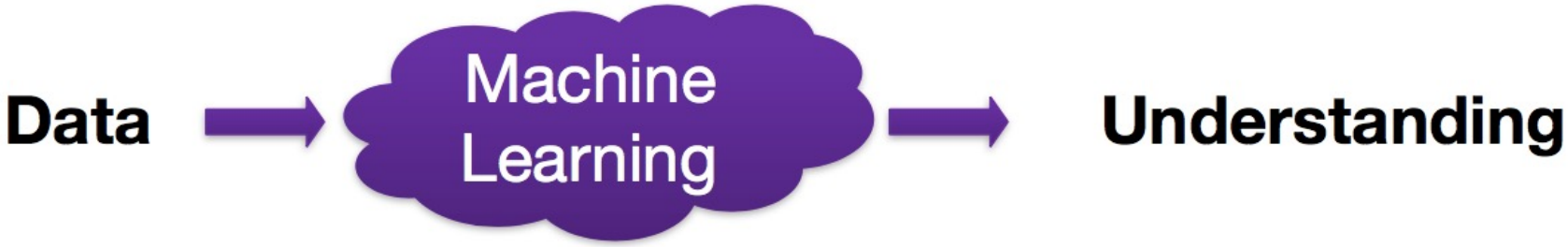

- **Machine Learning**




slide by Pedro Domingos, Tom Mitchel, Tom Dietterich

# What is Machine Learning?

- If you are a Scientist




- If you are an Engineer / Entrepreneur
  - Get lots of data
  - Machine Learning
  - ???
  - Profit!

slide by Dhruv Batra

# Why Study Machine Learning?

## Engineering Better Computing Systems

- Develop systems
  - too difficult/expensive to construct manually
  - because they require specific detailed skills/knowledge
  - ***knowledge engineering bottleneck***

- Develop systems
  - that adapt and customize themselves to individual users.
  - Personalized news or mail filter
  - Personalized tutoring

- Discover new knowledge from large databases
  - Medical text mining (e.g. migraines to calcium channel blockers to magnesium)
  - ***data mining***

slide by Dhruv Batra

# Why Study Machine Learning?

## Cognitive Science

- Computational studies of learning may help us understand learning in humans
  - and other biological organisms.

- Hebbian neural learning
  - “Neurons that fire together, wire together.”

slide by Dhruv Batra

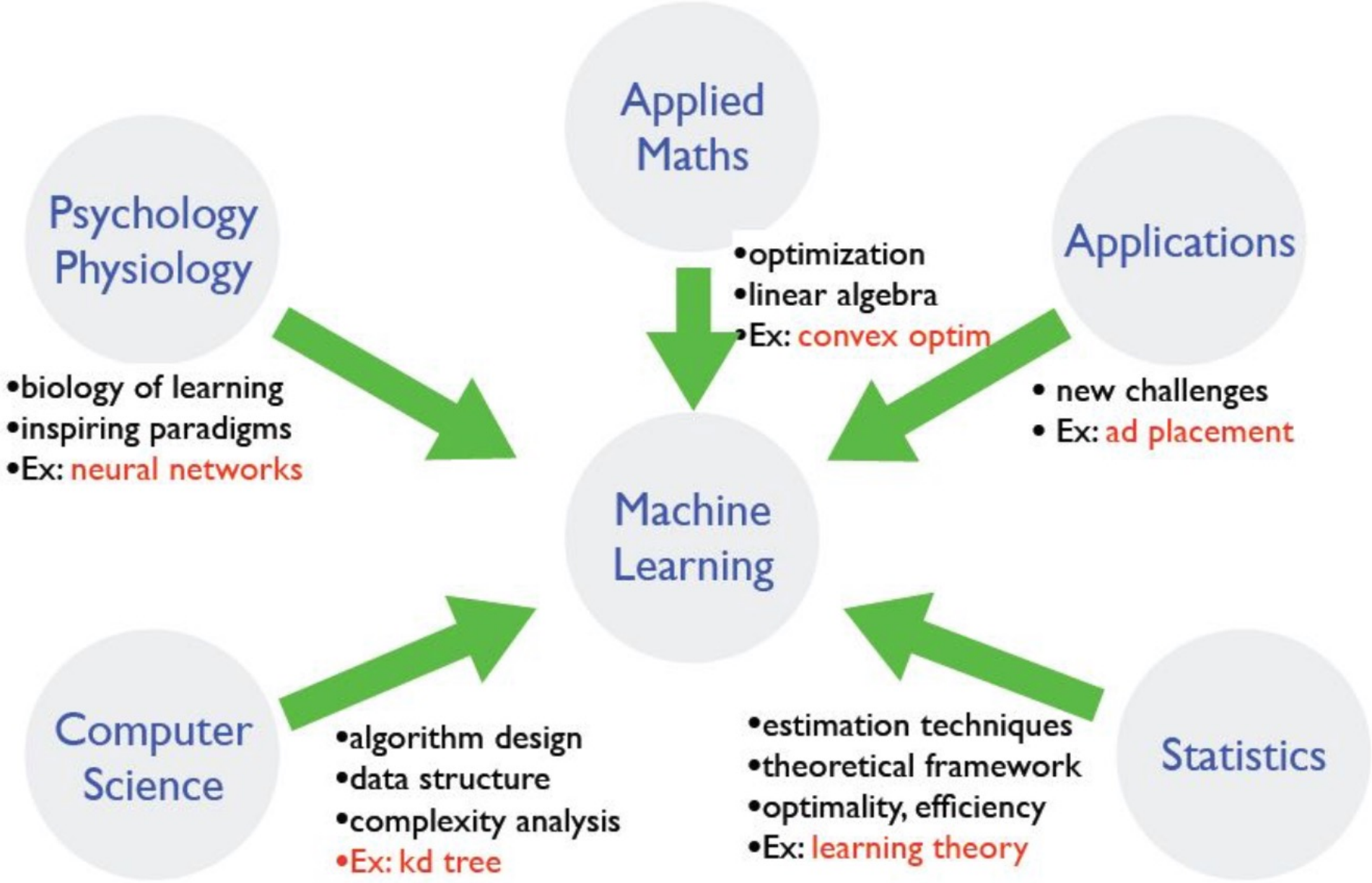
# Why Study Machine Learning?

## The Time is Ripe

- Algorithms
  - Many basic effective and efficient algorithms available.
- Data
  - Large amounts of on-line data available.
- Computing
  - Large amounts of computational resources available.

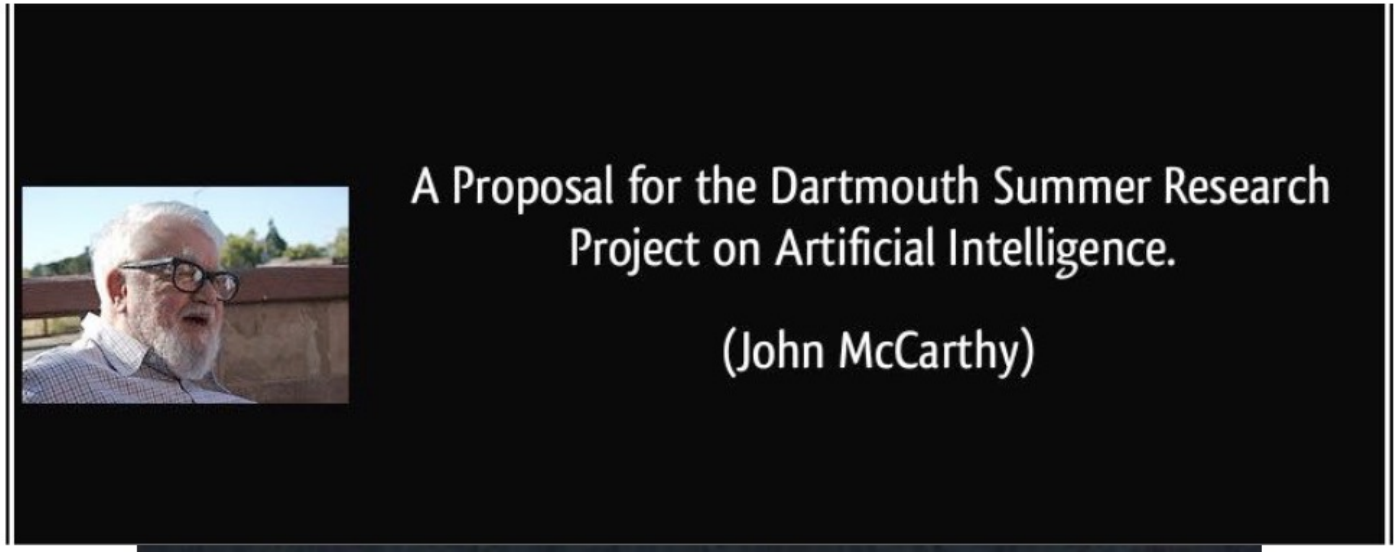
slide by Ray Mooney

# Where does ML fit in?




slide by Fei Sha

# A Brief History of AI






slide by Dhruv Batra

A Proposal for the

DARTMOUTH SUMMER RESEARCH PROJECT ON ARTIFICIAL INTELLIGENCE

June 17 - Aug. 16

We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer.

The following are some aspects of the artificial intelligence problem:

1) Automatic Computers

If a machine can do a job, then an automatic calculator can be programmed to simulate the machine. The speeds and memory capacities of present computers may be insufficient to simulate many of the higher functions of the human brain, but the major obstacle is not lack of machine capacity, but our inability to write programs taking full advantage of what we have.

2) How Can a Computer be Programmed to Use a Language

It may be speculated that a large part of human thought consists of manipulating words according to rules of reasoning



adopted from Dhruv Batra

# Why is AI hard?



Image Credit: http://karpathy.github.io/2012/10/22/state-of-computer-vision/

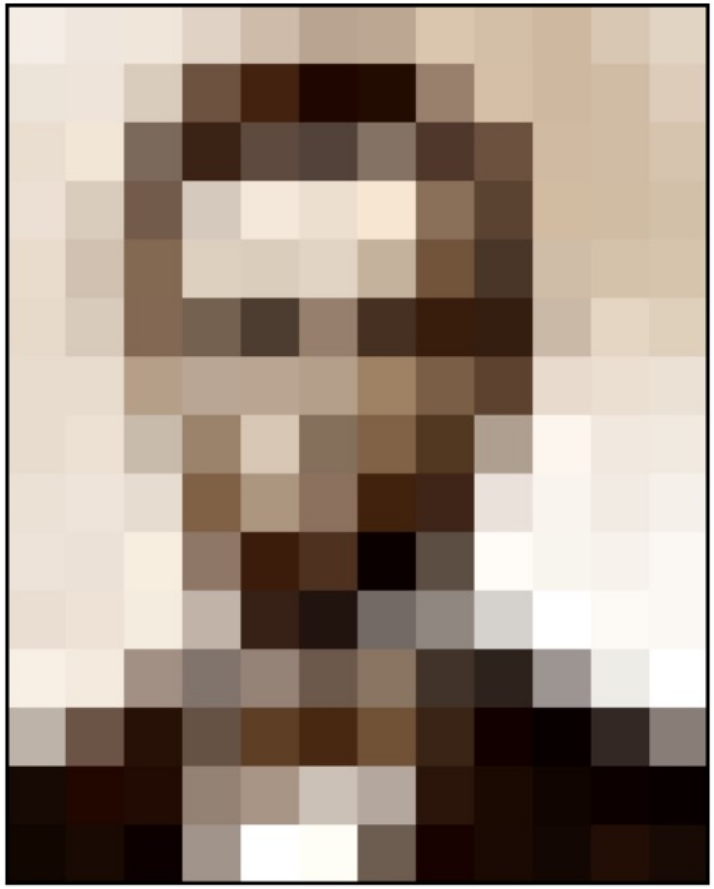
What would it take for a computer to understand this image as you do?

slide by Dhruv Batra

What would it take for a computer to understand this image as you or I do? I challenge you to think explicitly of all the pieces of knowledge that have to fall in place for it to make sense. Here is my short attempt:

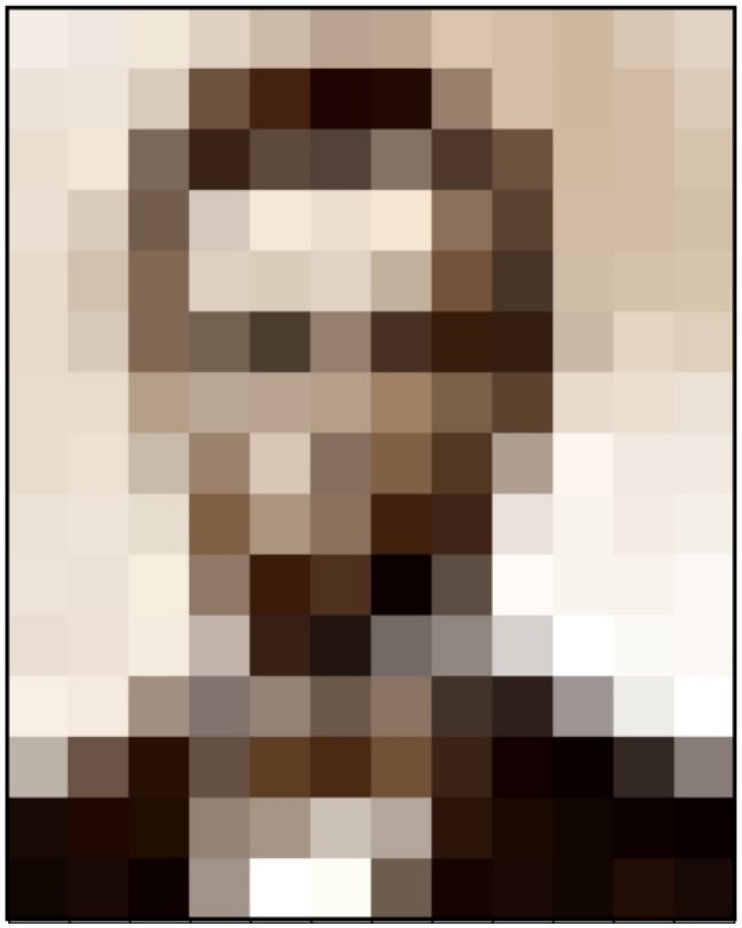
- You recognize it is an image of a bunch of people and you understand they are in a hallway
- You recognize that there are 3 mirrors in the scene so some of those people are "fake" replicas from different viewpoints.
- You recognize Obama from the few pixels that make up his face. It helps that he is in his suit and that he is surrounded by other people with suits.
- You recognize that there's a person standing on a scale, even though the scale occupies only very few white pixels that blend with the background. But, you've used the person's pose and knowledge of how people interact with objects to figure it out.
- You recognize that Obama has his foot positioned just slightly on top of the scale. Notice the language I'm using: It is in terms of the 3D structure of the scene, not the position of the leg in the 2D coordinate system of the image.
- You know how physics works: Obama is leaning in on the scale, which applies a force on it. Scale measures force that is applied on it, that's how it works => it will over-estimate the weight of the person standing on it.
- The person measuring his weight is not aware of Obama doing this. You derive this because you know his pose, you understand that the field of view of a person is finite, and you understand that he is not very likely to sense the slight push of Obama's foot.
- You understand that people are self-conscious about their weight. You also understand that he is reading off the scale measurement, and that shortly the over-estimated weight will confuse him because it will probably be much higher than what he expects. In other words, you reason about implications of the events that are about to unfold seconds after this photo was taken, and especially about the thoughts and how they will develop inside people's heads. You also reason about what pieces of information are available to people.
- There are people in the back who find the person's imminent confusion funny. In other words you are reasoning about state of mind of people, and their view of the state of mind of another person. That's getting frighteningly meta.
- Finally, the fact that the perpetrator here is the president makes it maybe even a little more funnier. You understand what actions are more or less likely to be undertaken by different people based on their status and identity.

# What humans see



slide by Larry Zitnick

# What computers see



slide by Larry Zitnick

# Why are things working today?

- More compute power
- More data
- Better algorithms/ models

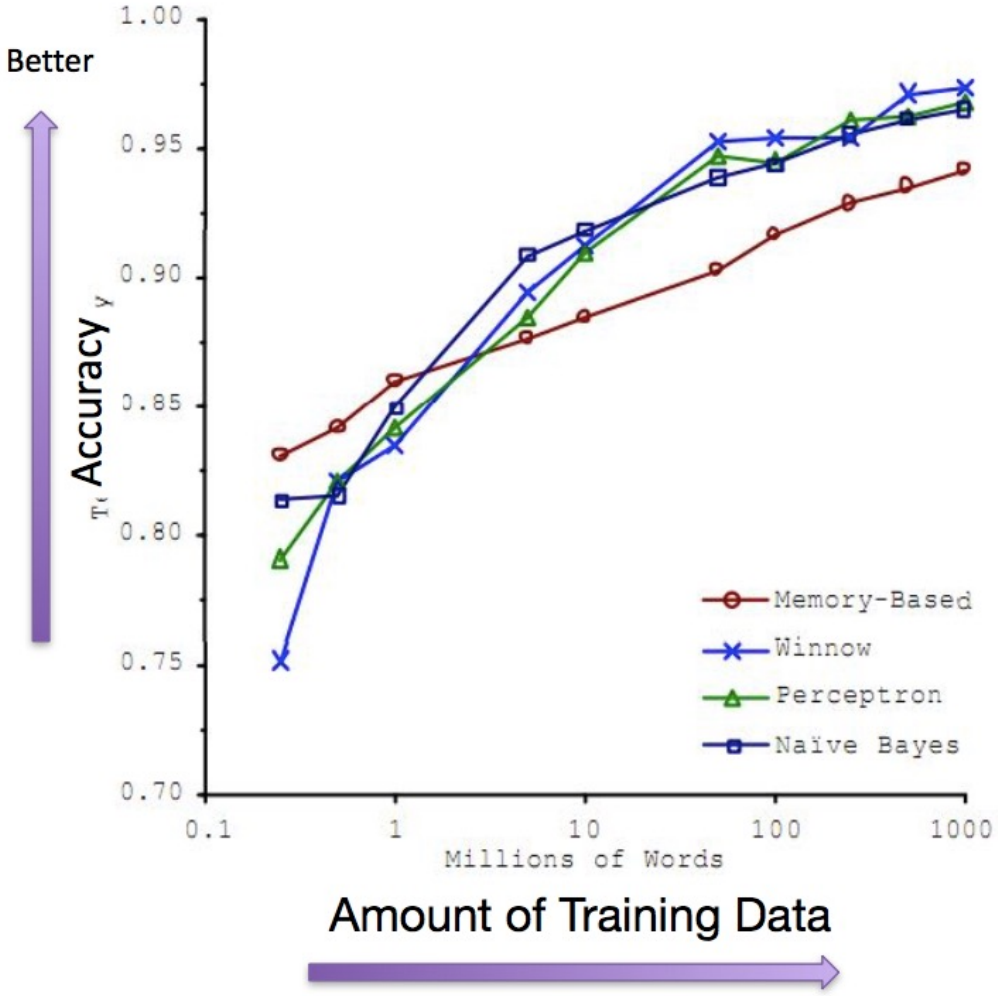



Figure Credit: Banko & Brill, 2011

slide by Dhruv Batra

# Machine Learning (by examples)

# Pose Estimation




slide by Alex Smola

# Learning to Predict Emergency C-Sections

[Sims et al., 2000]

Data:

| *Patient103* time=1 | *Patient103* time=2 | ... | *Patient103* time=n |
|---|---|---|---|
| Age: 23 | Age: 23 | | Age: 23 |
| FirstPregnancy: no | FirstPregnancy: no | | FirstPregnancy: no |
| Anemia: no | Anemia: no | | Anemia: no |
| Diabetes: no | Diabetes: YES | | Diabetes: no |
| PreviousPrematureBirth: no | PreviousPrematureBirth: no | | PreviousPrematureBirth: no |
| Ultrasound: ? | Ultrasound: abnormal | | Ultrasound: ? |
| Elective C–Section: ? | Elective C–Section: no | | Elective C–Section: no |
| Emergency C–Section: ? | Emergency C–Section: ? | | **Emergency C–Section: Yes** |
| ... | ... | | ... |

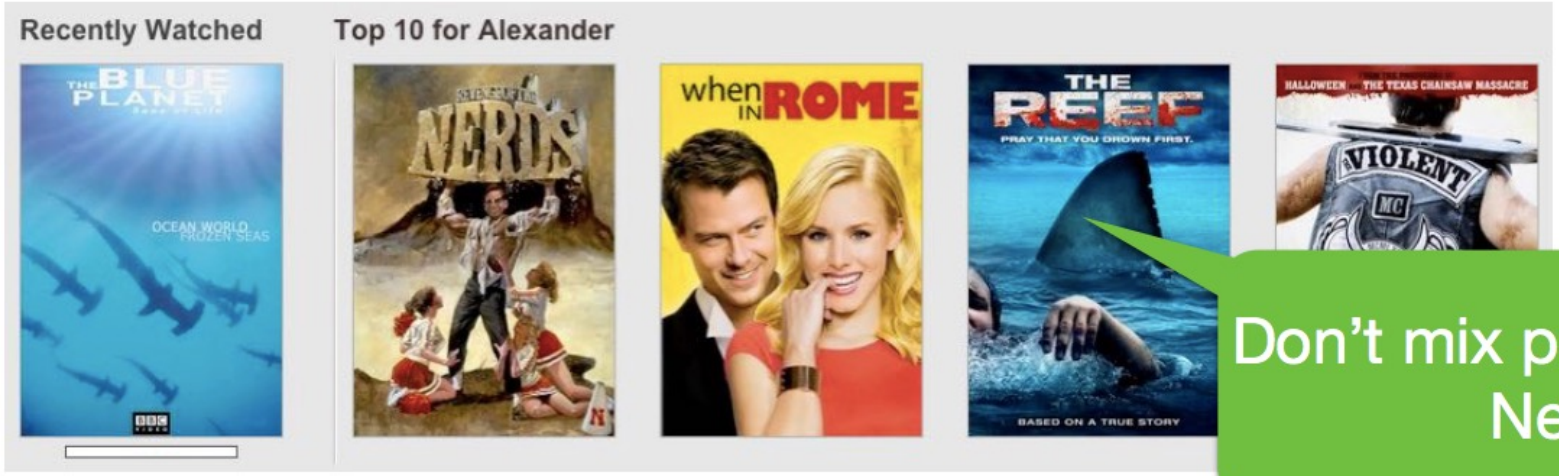
9714 patient records, each with 215 features

One of 18 learned rules:

```
If   No previous vaginal delivery, and
     Abnormal 2nd Trimester Ultrasound, and
     Malpresentation at admission
Then Probability of Emergency C-Section is 0.6

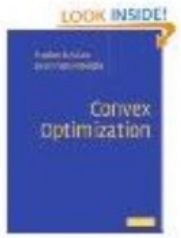
 Over training data: 26/41 = .63,
 Over test data: 12/20 = .60
```

# Collaborative Filtering




Don't mix preferences on Netflix!

## Customers Who Bought This Item Also Bought




Convex Optimization by Stephen Boyd

(11)

$65.78




Point Processes (Chapman & Hall / CRC Monographs on S... by D.R. Cox

$125.47



Probabilistic Graphical Models: Principles and T... by Daphne Koller

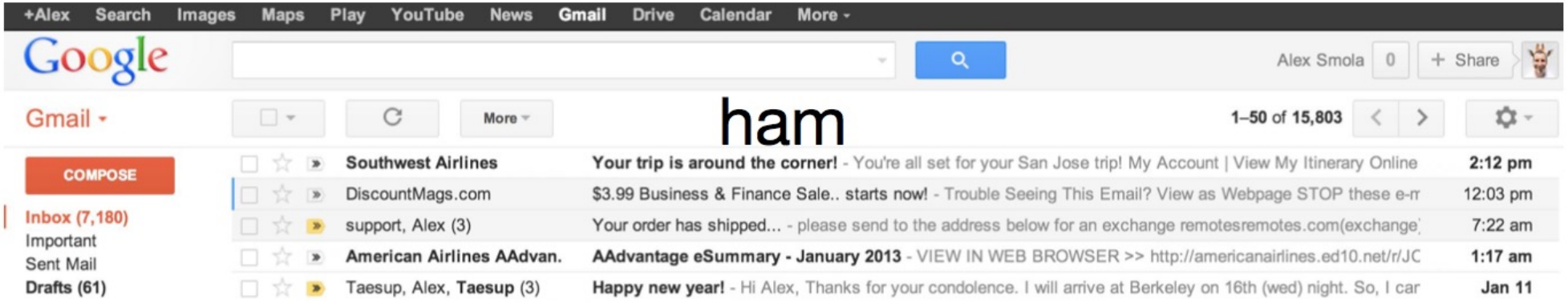
(5)

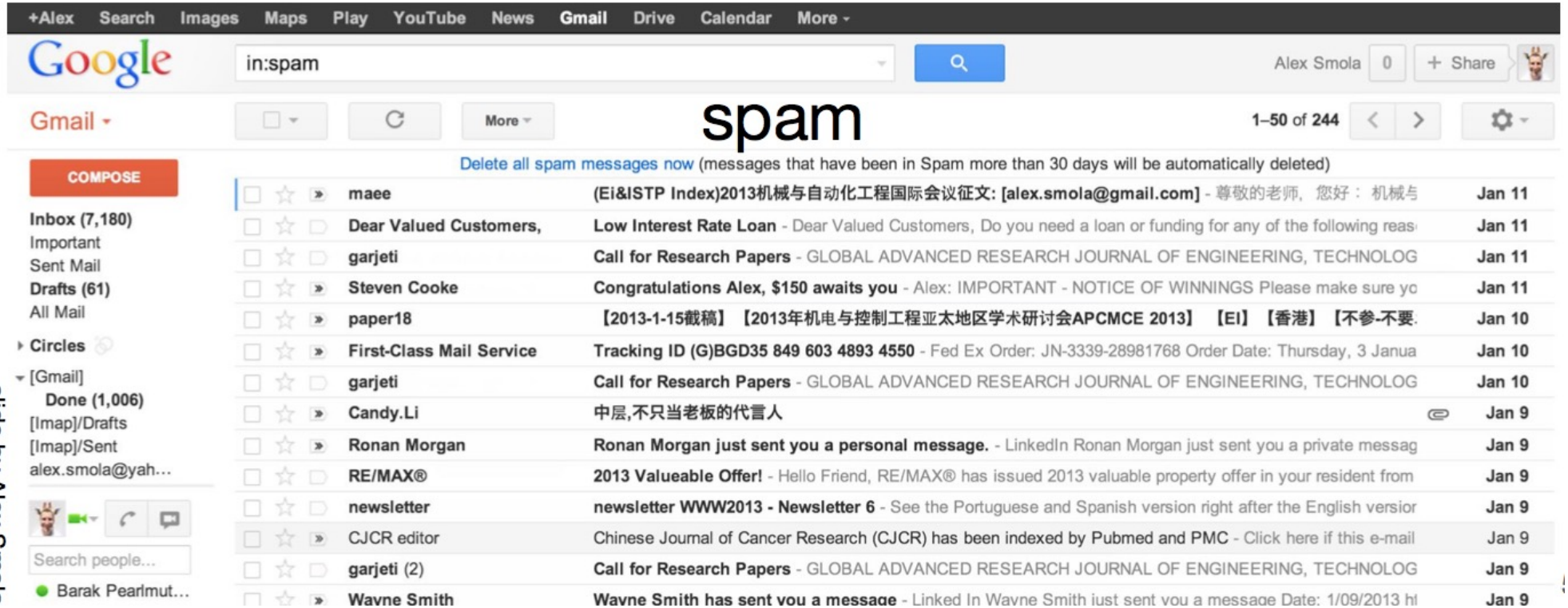
$71.52

Amazon books

slide by Alex Sr

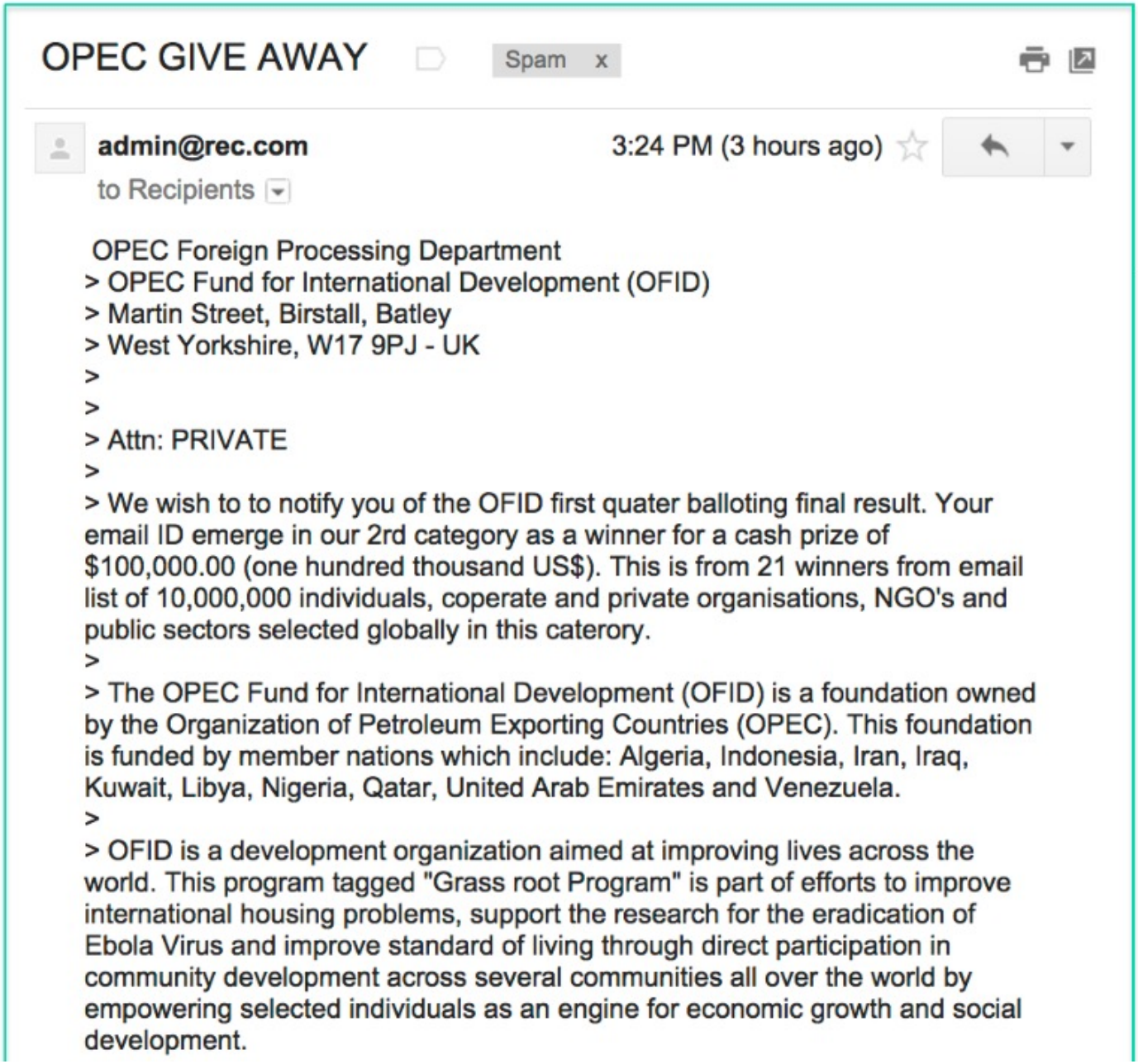
# Spam Filtering




slide by Alex Smola

# Learning to classify text documents



OPEC GIVE AWAY

Spam x

**admin@rec.com** 3:24 PM (3 hours ago)

to Recipients

OPEC Foreign Processing Department
> OPEC Fund for International Development (OFID)
> Martin Street, Birstall, Batley
> West Yorkshire, W17 9PJ - UK
>
>
> Attn: PRIVATE
>
> We wish to to notify you of the OFID first quater balloting final result. Your email ID emerge in our 2rd category as a winner for a cash prize of $100,000.00 (one hundred thousand US$). This is from 21 winners from email list of 10,000,000 individuals, coperate and private organisations, NGO's and public sectors selected globally in this caterory.
>
> The OPEC Fund for International Development (OFID) is a foundation owned by the Organization of Petroleum Exporting Countries (OPEC). This foundation is funded by member nations which include: Algeria, Indonesia, Iran, Iraq, Kuwait, Libya, Nigeria, Qatar, United Arab Emirates and Venezuela.
>
> OFID is a development organization aimed at improving lives across the world. This program tagged "Grass root Program" is part of efforts to improve international housing problems, support the research for the eradication of Ebola Virus and improve standard of living through direct participation in community development across several communities all over the world by empowering selected individuals as an engine for economic growth and social development.

→

spam

vs

not spam

# Self-Driving Cars

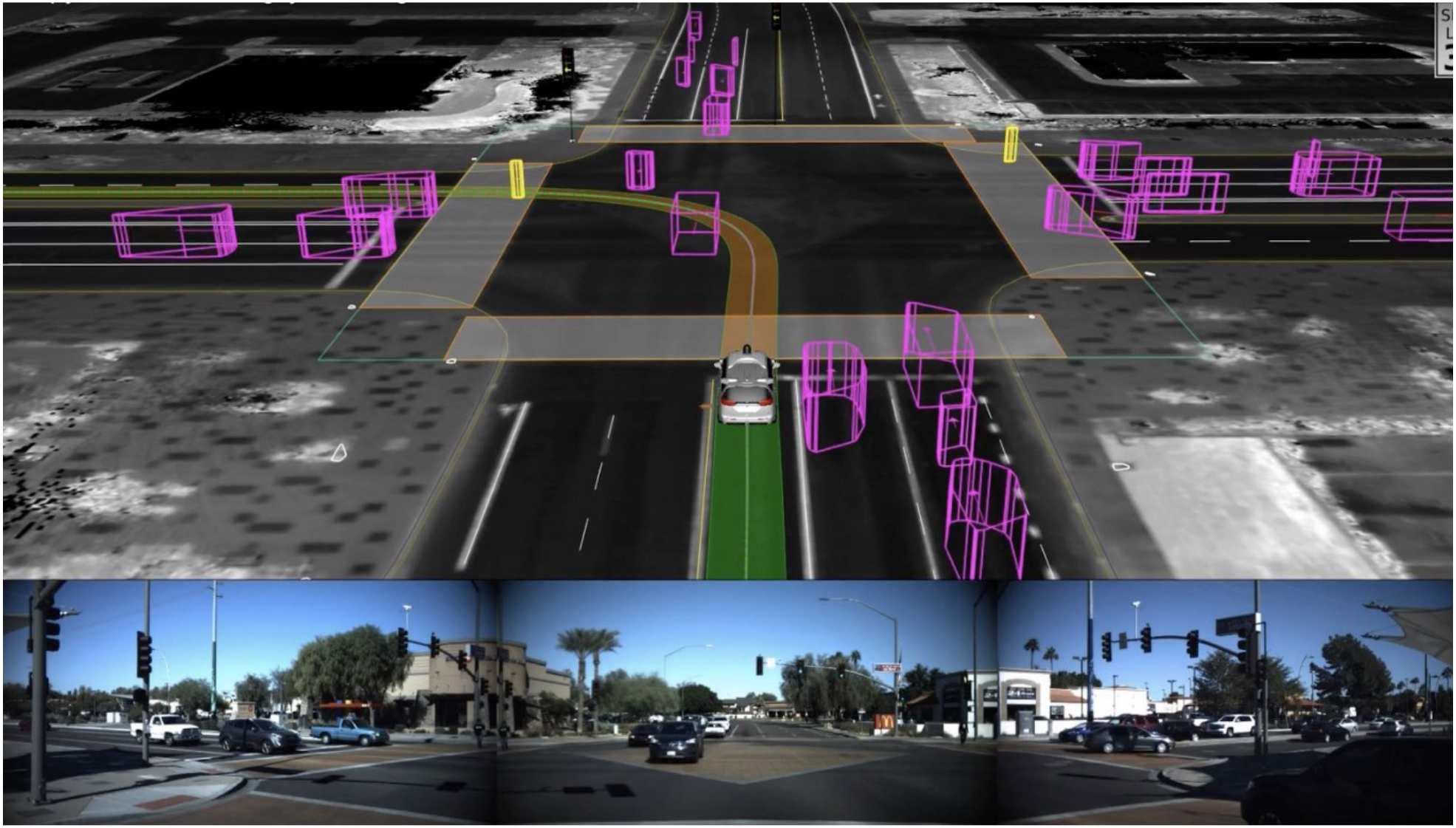
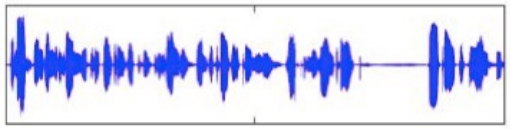
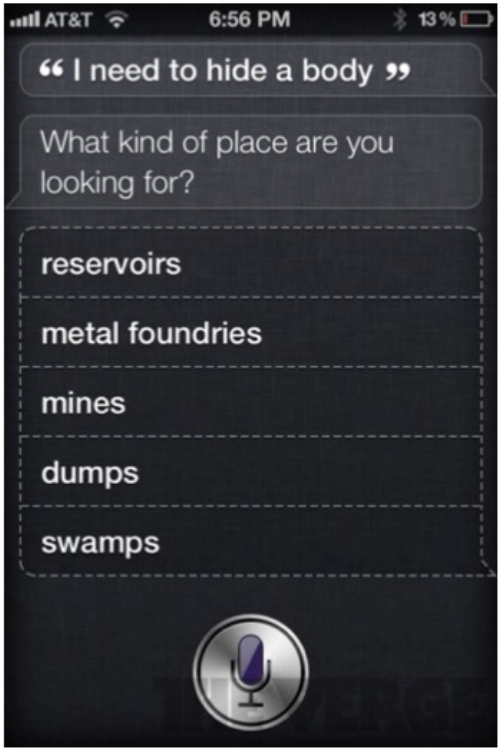


Image: https://medium.com/waymo/simulation-how-one-flashing-yellow-light-turns-into-thousands-of-hours-of-experience-a7a1cb475565

# Speech Recognition

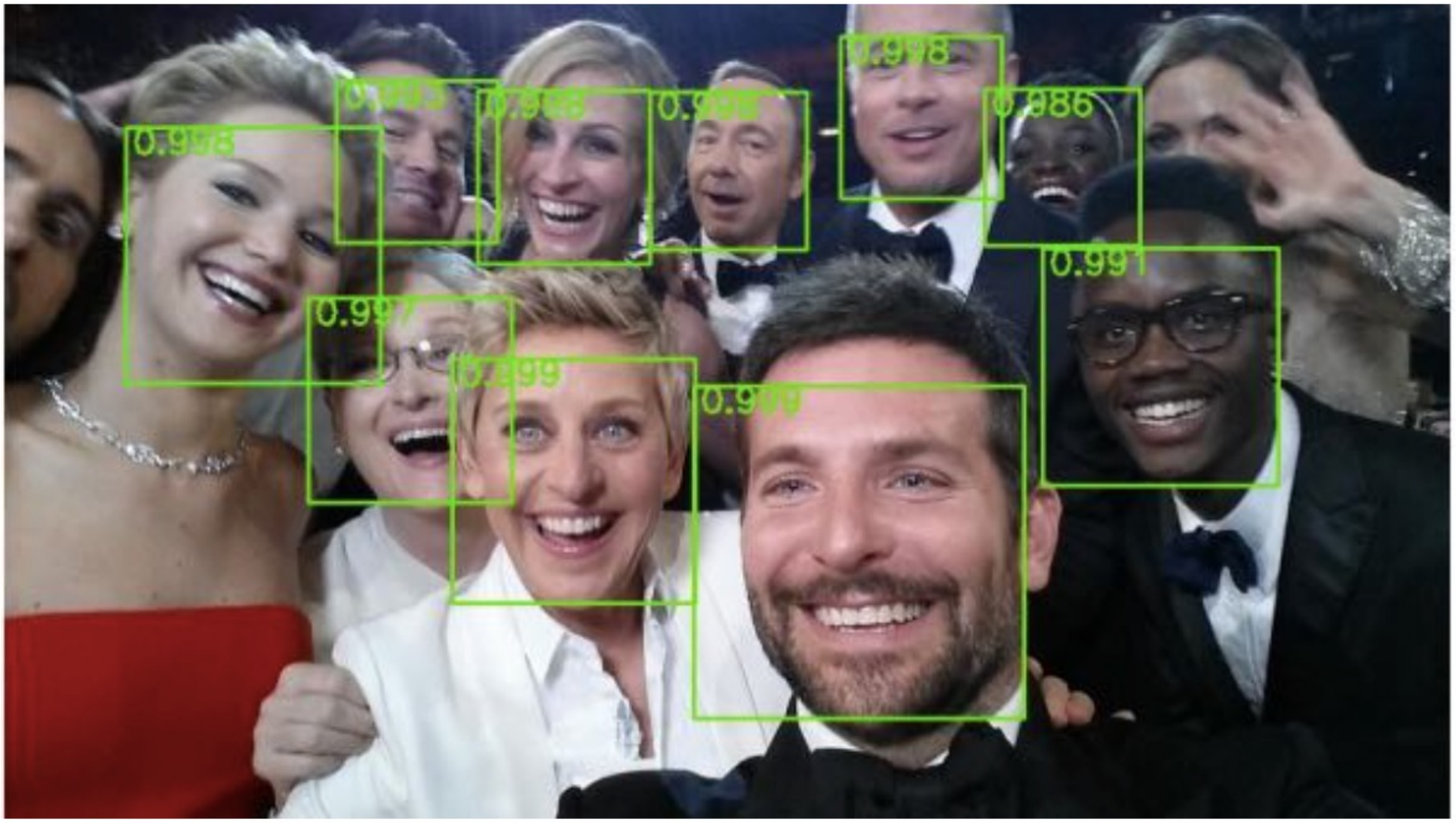
Given an audio waveform, robustly extract & recognize any spoken words





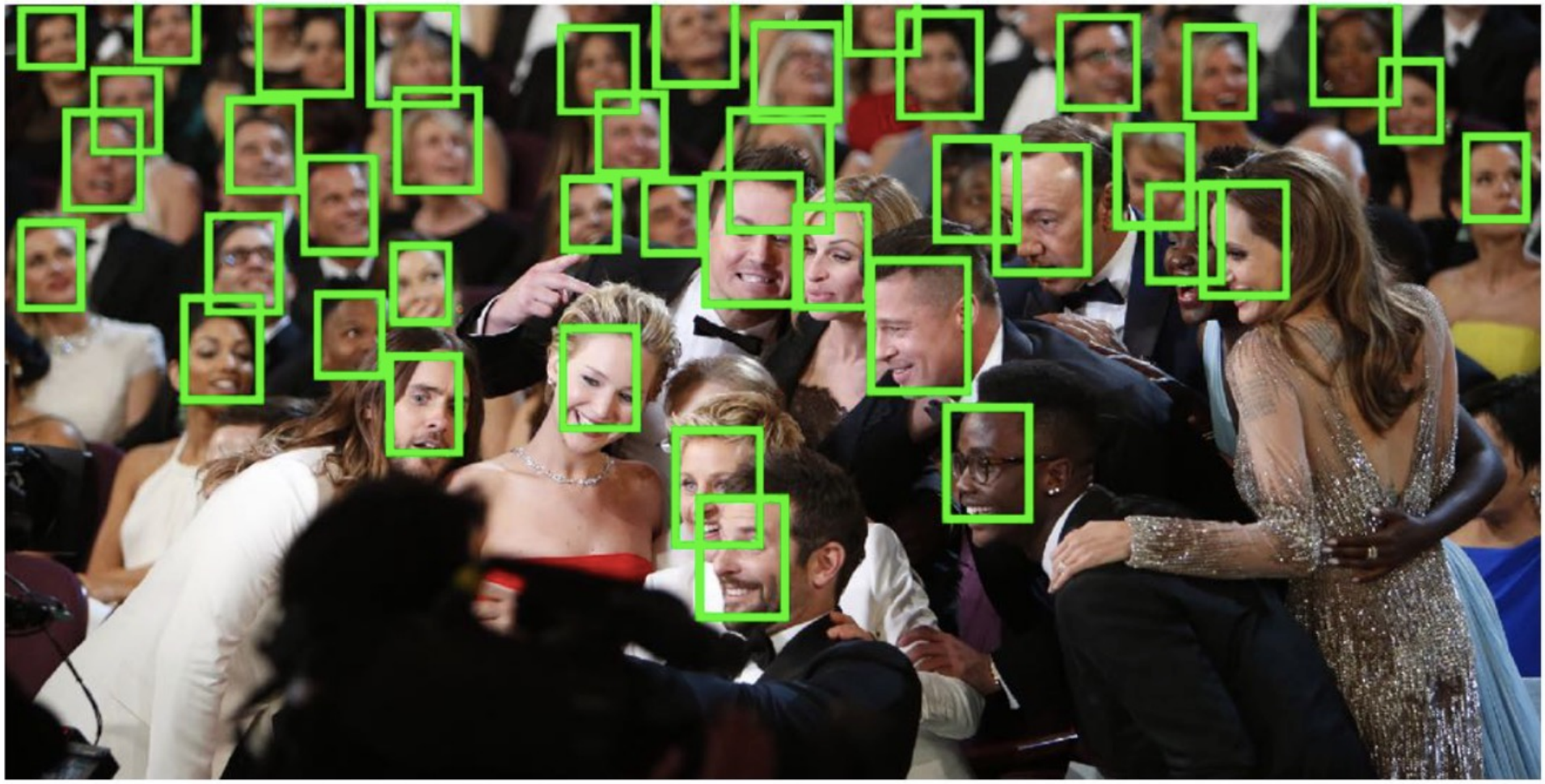

- Statistical models can be used to
  - Provide greater robustness to noise
  - Adapt to accent of different speakers
  - Learn from training

# Face Detection



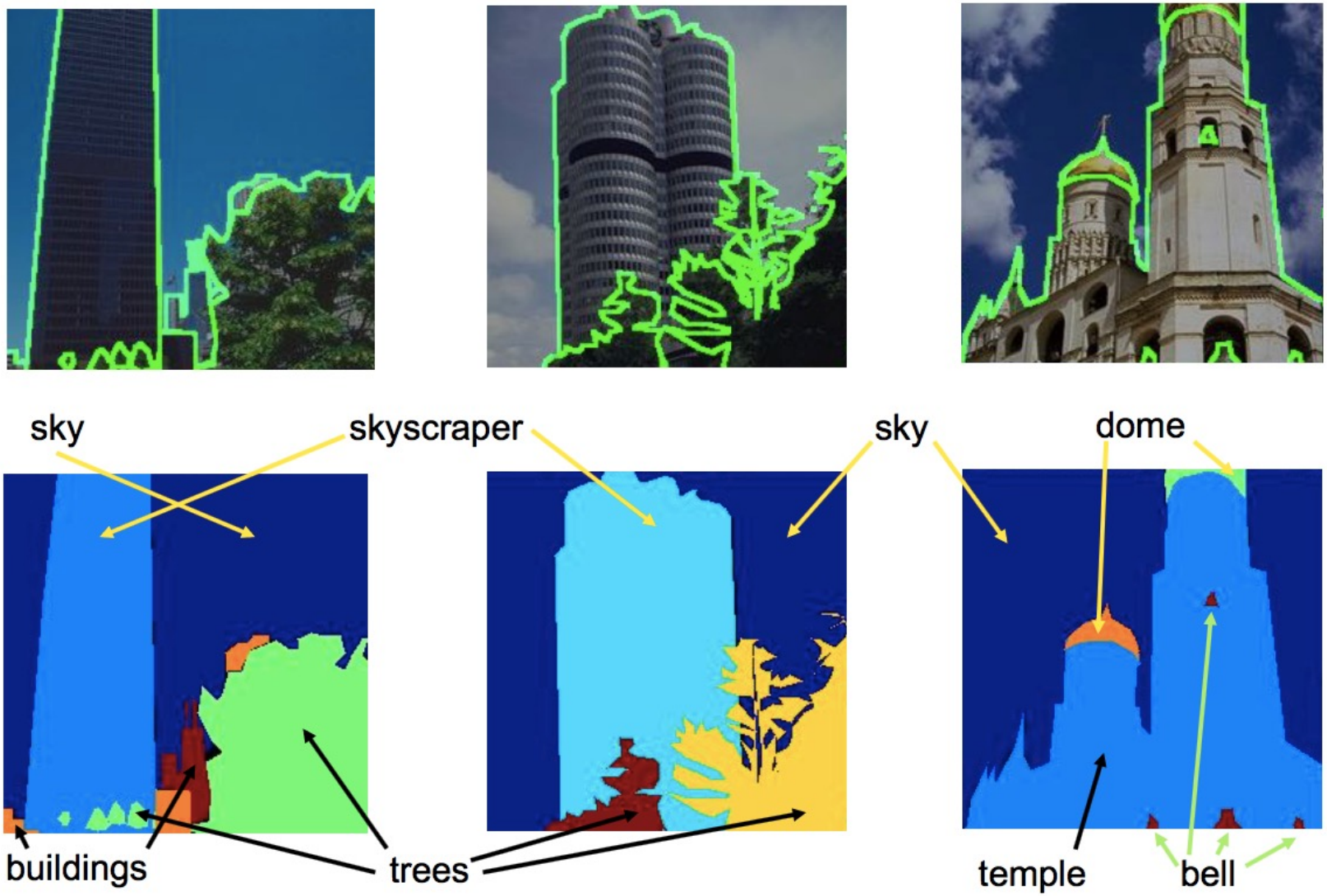

Sudhakar et al., Multi-view Face Detection Using Deep Convolutional Neural Networks, 2015

# Face Detection



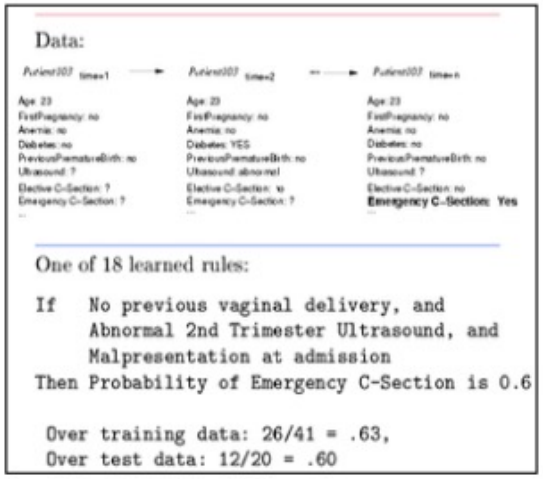
Yang et al., From Facial Parts Responses to Face Detection: A Deep Learning Approach, ICCV 2015

# Visual Scene Understanding

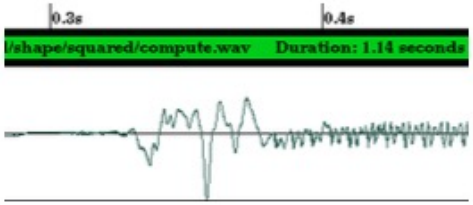



slide by Eric Sudderth

# Machine Learning - Practice




Mining Databases




Speech Recognition



Object recognition



Control learning

Text analysis

**Peter H. van Oppen** , Chairman of the Board & Chief Executive Officer
Mr. van Oppen has served as Chairman of the board and chief executive officer of ADIC since its acquisition by Interpoint in 1994 and a director of ADIC since 1986. Until its acquisition by Crane Co. in October 1996, Mr. van Oppen served as Chairman of the board of directors, president and chief executive officer of Interpoint . Prior to 1985, Mr. van Oppen worked as a consulting manager at Price Waterhouse LLP and at Bain & Company in Boston and London. He has additional experience in medical electronics and venture capital. Mr. van Oppen also serves as a director of Seattle FilmWorks Inc. and Spacelabs Medical, Inc.. He holds a B.A. from Whitman College and an M.B.A. from Harvard Business School, where he was a Baker Scholar.

- Support Vector Machines
- Bayesian networks
- Hidden Markov models
- Deep neural networks
- Reinforcement learning
- ....

# Machine Learning - Theory

PAC Learning Theory
(supervised concept learning)

# examples (*m*)

representational complexity (*H*)

error rate (ε)

failure probability (δ)

$$m \geq \frac{1}{\epsilon}(\ln |H| + \ln(1/\delta))$$

Other theories for

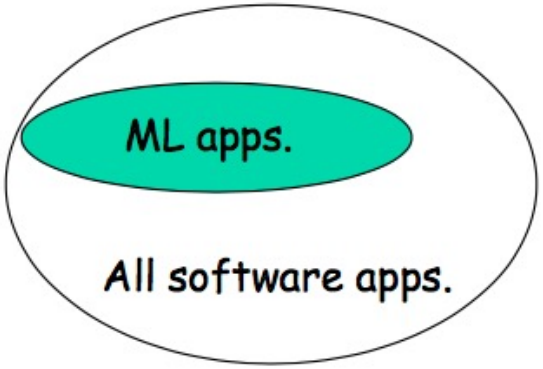
- Reinforcement skill learning
- Semi-supervised learning
- Active student querying
- …

… also relating:

- # of mistakes during learning
- learner's query strategy
- convergence rate
- asymptotic performance
- bias, variance

# Machine Learning in Computer Science

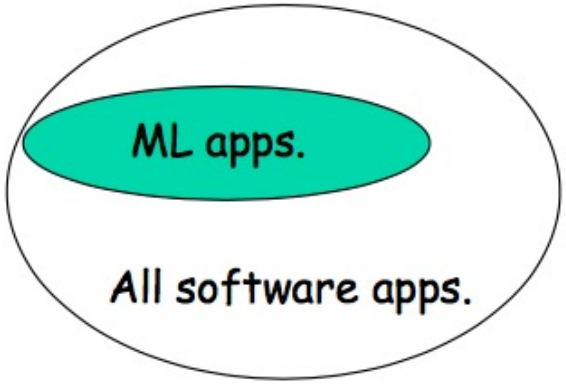
- Machine learning already the preferred approach to
  - Speech recognition, Natural language processing
  - Computer vision
  - Medical outcomes analysis
  - Robot control
  - …




- This ML niche is growing (why?)

# Machine Learning in Computer Science

- Machine learning already the preferred approach to
  - Speech recognition, Natural language processing
  - Computer vision
  - Medical outcomes analysis
  - Robot control
  - …




- This ML niche is growing
  - Improved machine learning algorithms
  - Increased volume of online data
  - Increased demand for self-customizing software

# Programming with Data

- Want adaptive robust and fault tolerant systems
- Rule-based implementation is (often)
  - difficult (for the programmer)
  - brittle (can miss many edge-cases)
  - becomes a nightmare to maintain explicitly
  - often doesn't work too well (e.g. OCR)

- Usually easy to obtain examples of what we want
  IF x THEN DO y
- Collect many pairs $(x_i, y_i)$
- Estimate function f such that $f(x_i) = y_i$ (supervised learning)
- Detect patterns in data (unsupervised learning)

slide by Mehryar Mohri

# Objectives of Machine Learning

- Algorithms: design of efficient, accurate, and general learning algorithms to
  - deal with large-scale problems.
  - make accurate predictions (unseen examples).
  - handle a variety of different learning problems.
- Theoretical questions:
  - what can be learned? Under what conditions?
  - what learning guarantees can be given?
  - what is the algorithmic complexity?

slide by Mehryar Mohri

# Definitions and Terminology

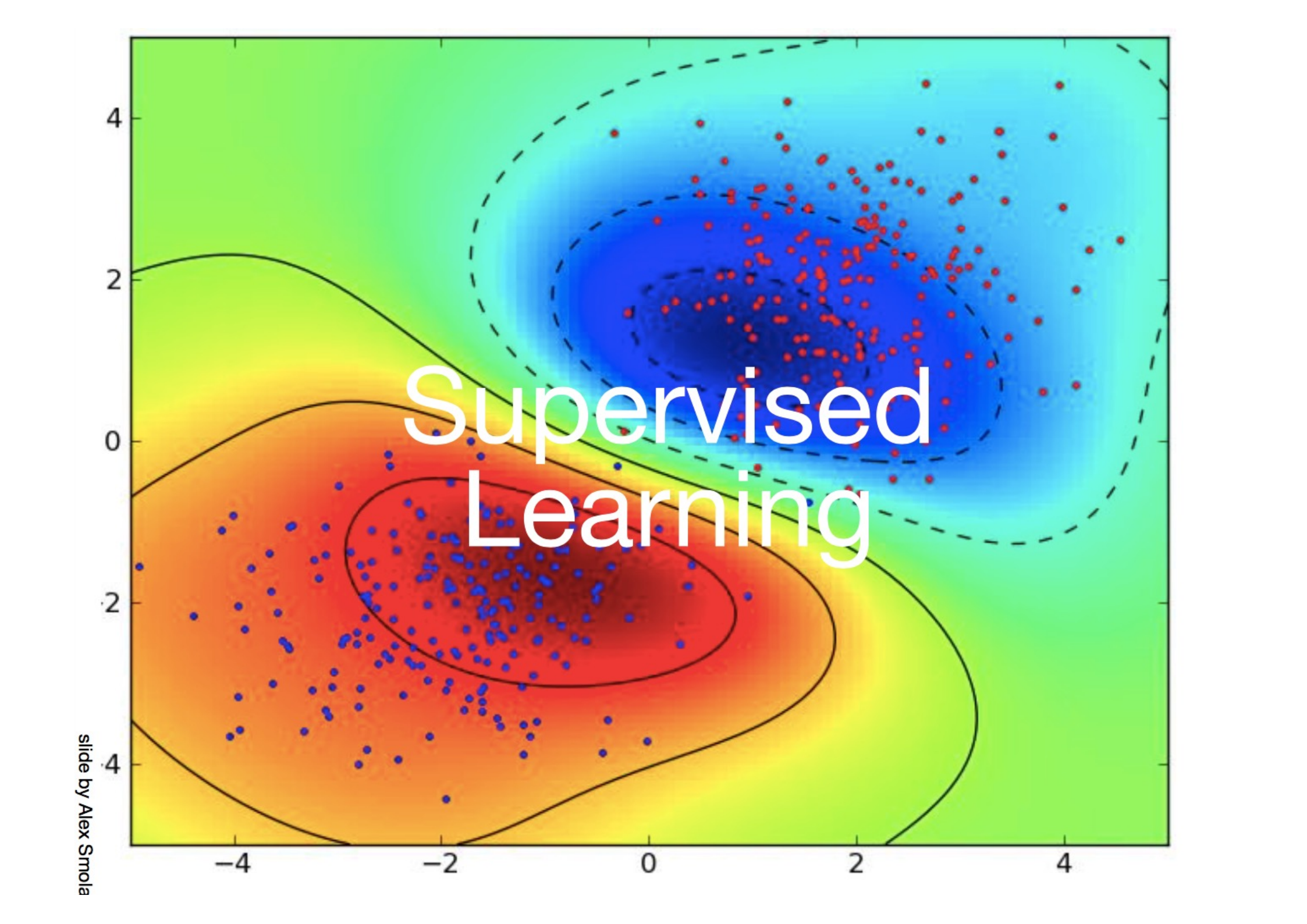
- Example: an object, instance of the data used.
- Features: the set of attributes, often represented as a vector, associated to an example (e.g., height and weight for gender prediction).
- Labels: in classification, category associated to an object (e.g., positive or negative in binary classification); in regression real value.
- Training data: data used for training learning algorithm (often labeled data).

slide by Mehryar Mohri

# Definitions and Terminology (cont'd.)

- Test data: data used for testing learning algorithm (unlabeled data).
- Unsupervised learning: no labeled data.
- Supervised learning: uses labeled data.
- Weakly or semi-supervised learning: intermediate scenarios.
- Reinforcement learning: rewards from sequence of action.

slide by Mehryar Mohri




slide by Alex Smola

# Supervised Learning

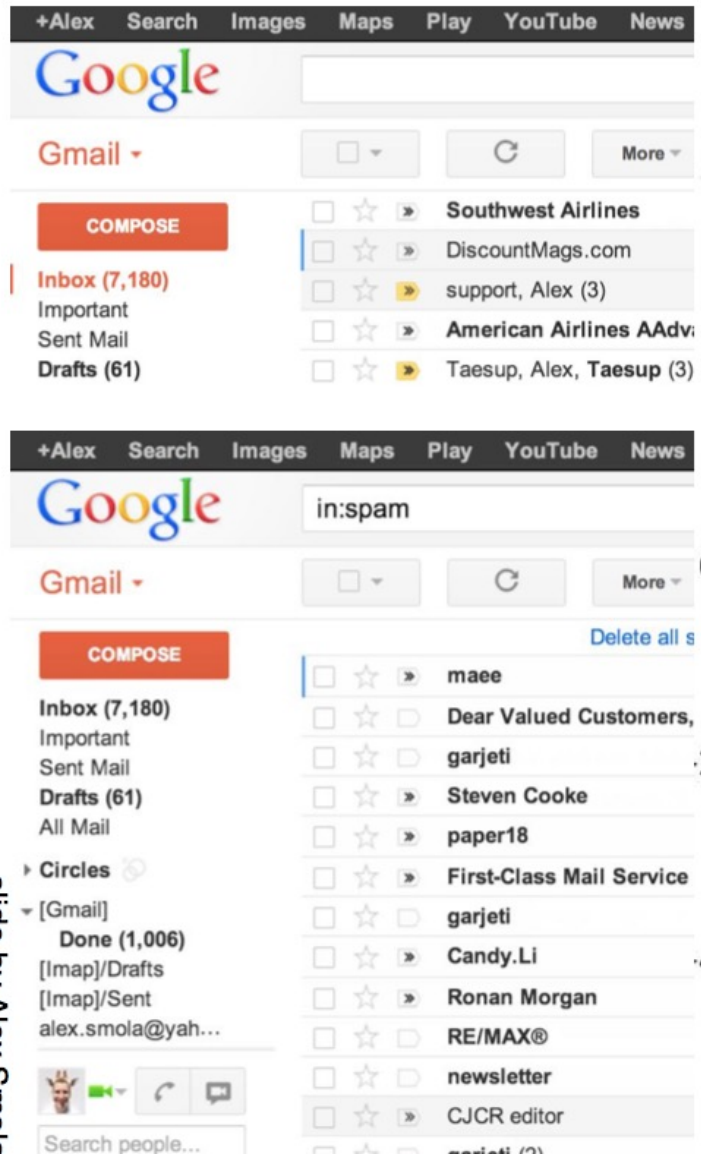
- **Binary classification**
  Given x find y in {-1, 1}
- **Multicategory classification**
  Given x find y in {1, ... k}
- **Regression**
  Given x find y in R (or $R^d$)
- **Sequence annotation**
  Given sequence $x_1$ ... $x_l$ find $y_1$ ... $y_l$
- **Hierarchical Categorization (Ontology)**
  Given x find a point in the hierarchy of y (e.g. a tree)
- **Prediction**
  Given $x_t$ and $y_{t-1}$ ... $y_1$ find $y_t$

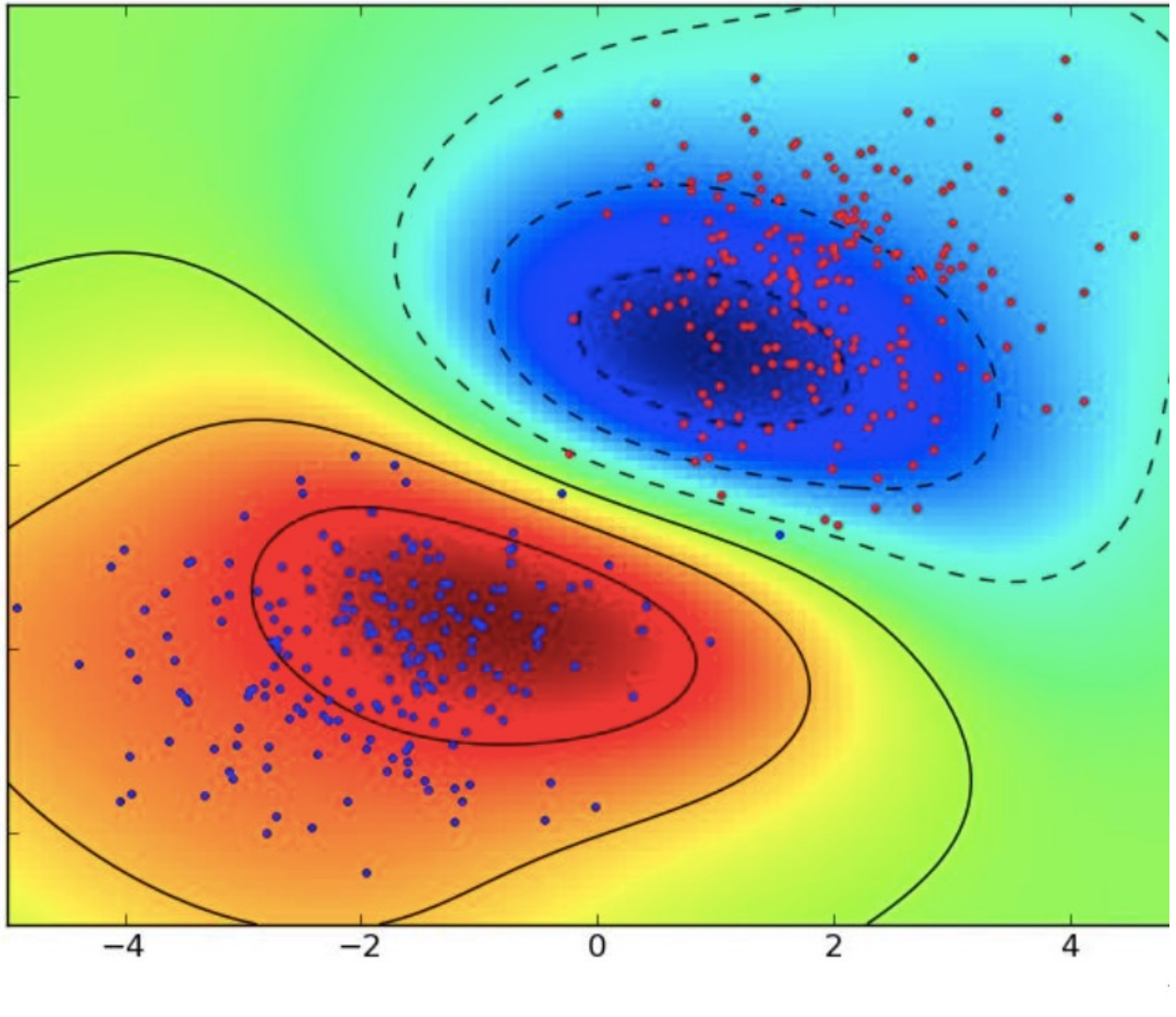
often with loss

$$l(y, f(x))$$

slide by Alex Smola

# Binary Classification




slide by Alex Smola

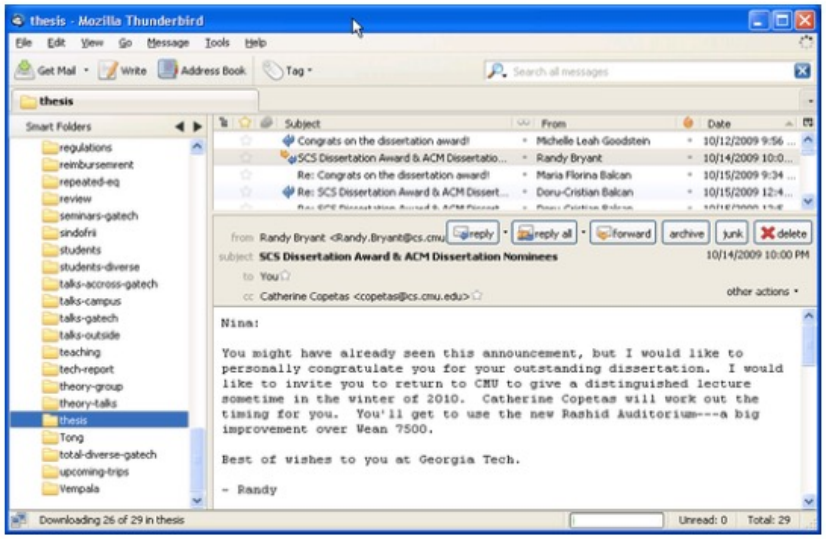
# Supervised Classification. Example: Spam Detection

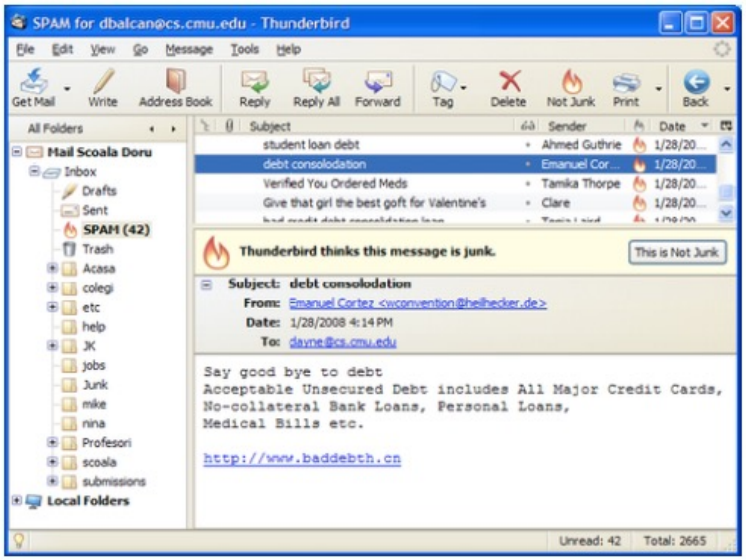
Decide which emails are spam and which are important.

Supervised classification

Not spam

spam




**Goal: use emails seen so far to produce good prediction rule for future data.**

# Supervised Classification. Example: Spam Detection

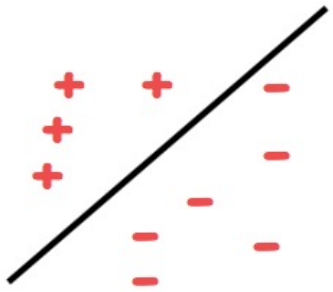
Represent each message by features. (e.g., keywords, spelling, etc.)

| | "money" | "pills" | "Mr." | bad spelling | known-sender | spam? | |
|---|---|---|---|---|---|---|---|
| | Y | N | Y | Y | N | Y | |
| | N | N | N | Y | Y | N | |
| | N | Y | N | N | N | Y | |
| example | Y | N | N | N | Y | N | label |
| | N | N | Y | N | Y | N | |
| | Y | N | N | Y | N | Y | |
| | N | N | Y | N | N | N | |

Reasonable RULES:

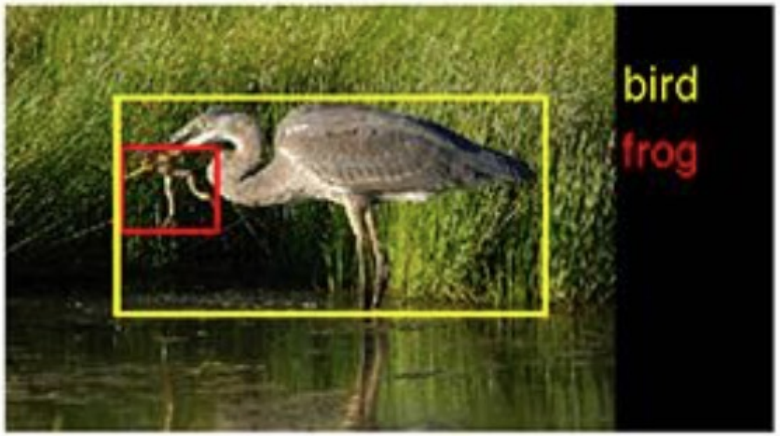
Predict SPAM if unknown AND (money OR pills)

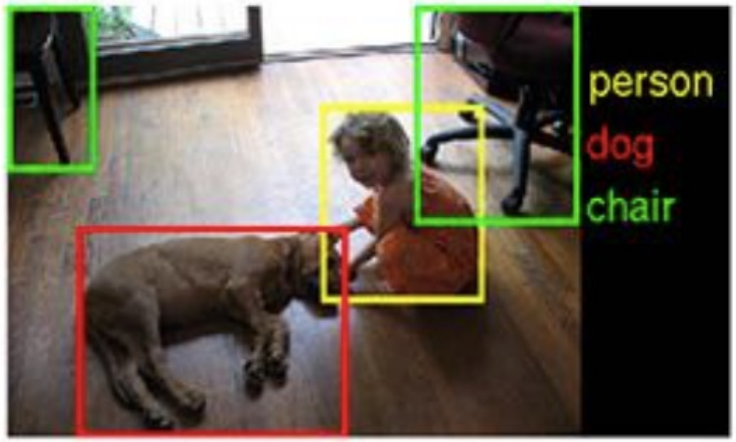
Predict SPAM if 2money + 3pills –5 known > 0



Linearly separable

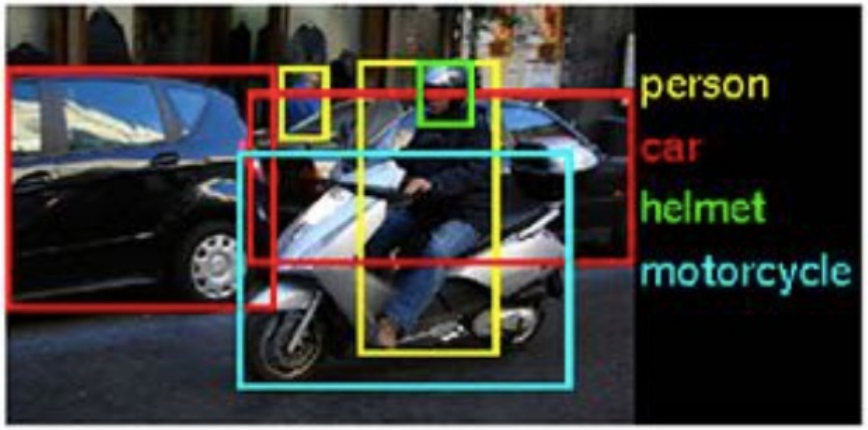
# Multiclass Classification + Annotation



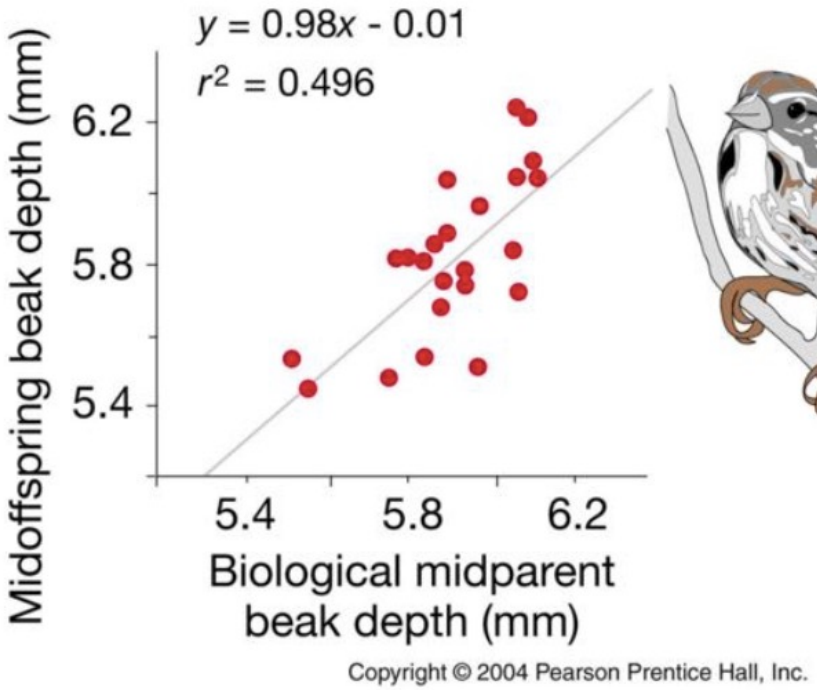







slide by Alex Smola

# Regression

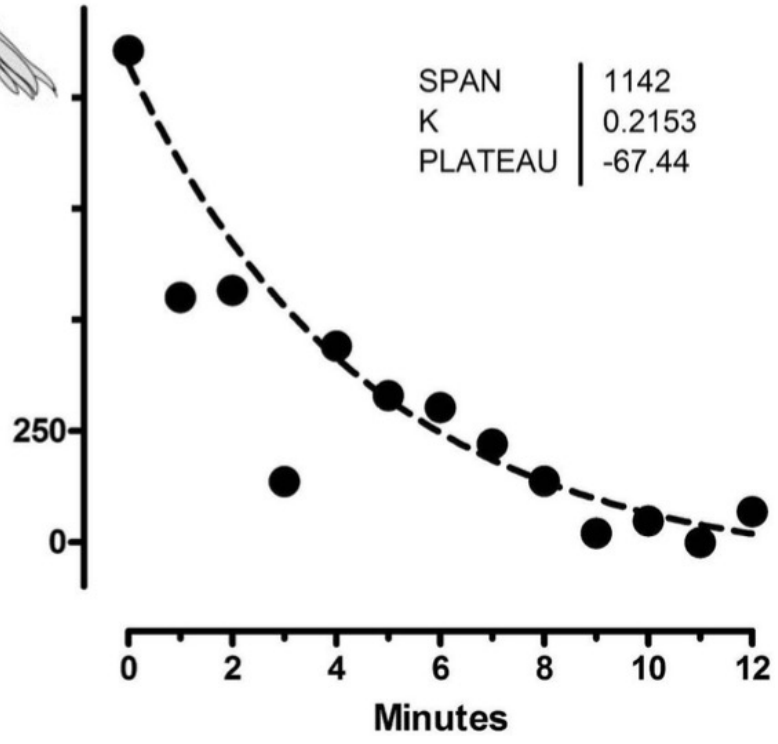


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linear



nonlinear



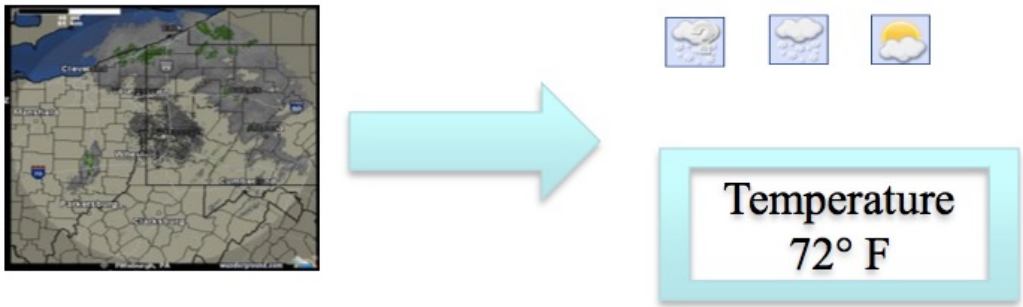

slide by Alex Smola

# Regression. Predicting a numeric value

## Stock market




## Weather prediction




Predict the temperature at any given location

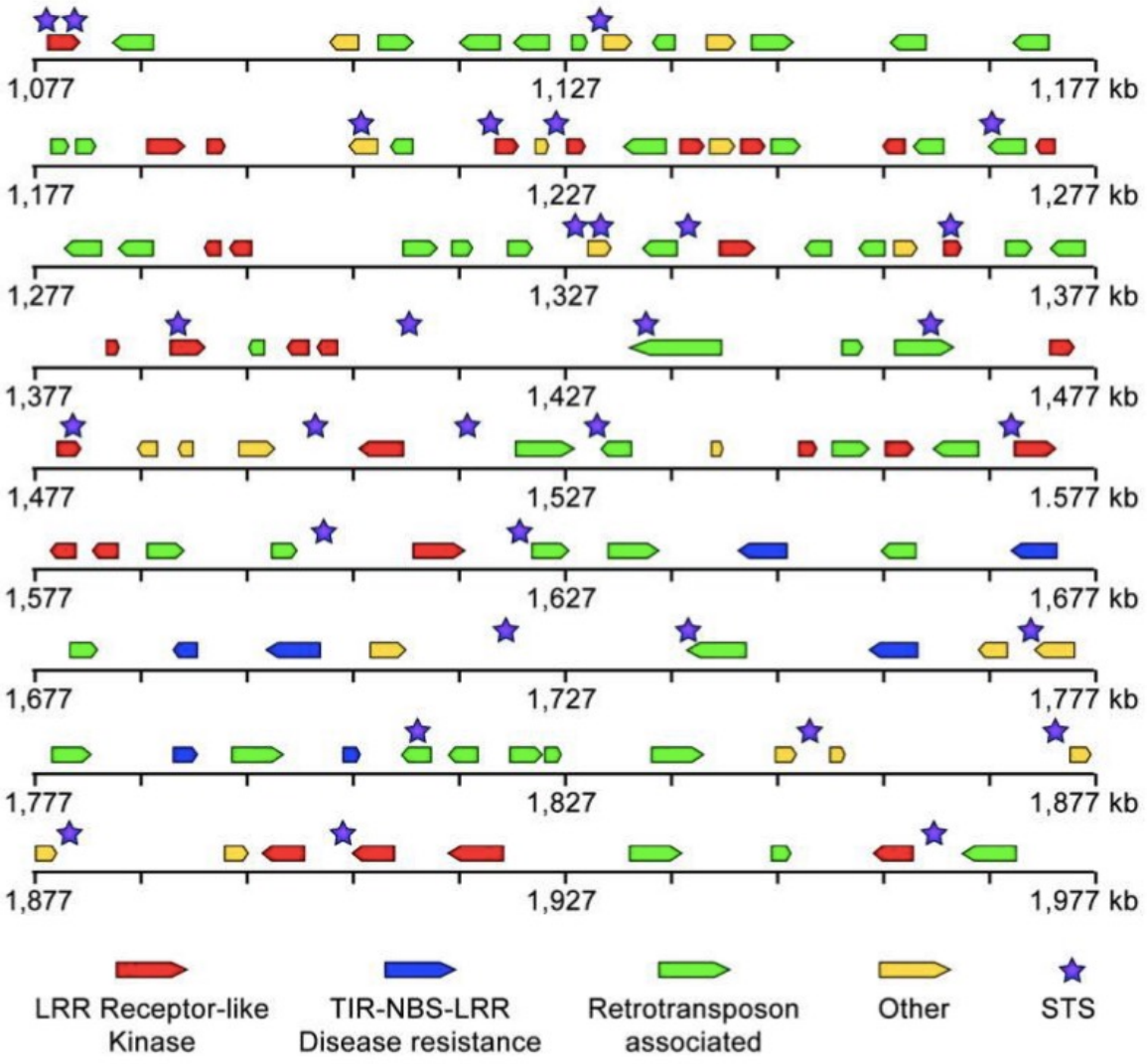
# Prediction




slide by Alex Smola

tomorrow's stock price

# Sequence Annotation



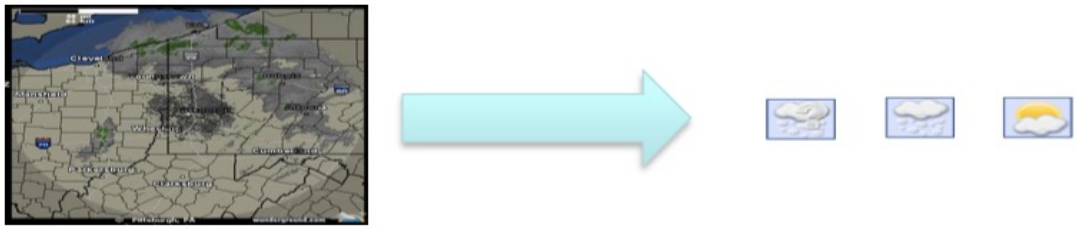

given sequence

gene finding
speech recognition
activity segmentation
named entities

slide by Alex Sm

# Supervised Classification. Many other examples

- Weather prediction



- Medicine:
  - diagnose a disease
    - input: from symptoms, lab measurements, test results, DNA tests, …
    - output: one of set of possible diseases, or "none of the above"
    - examples: audiology, thyroid cancer, diabetes, …
      - or: response to chemo drug X
      - or: will patient be re-admitted soon?

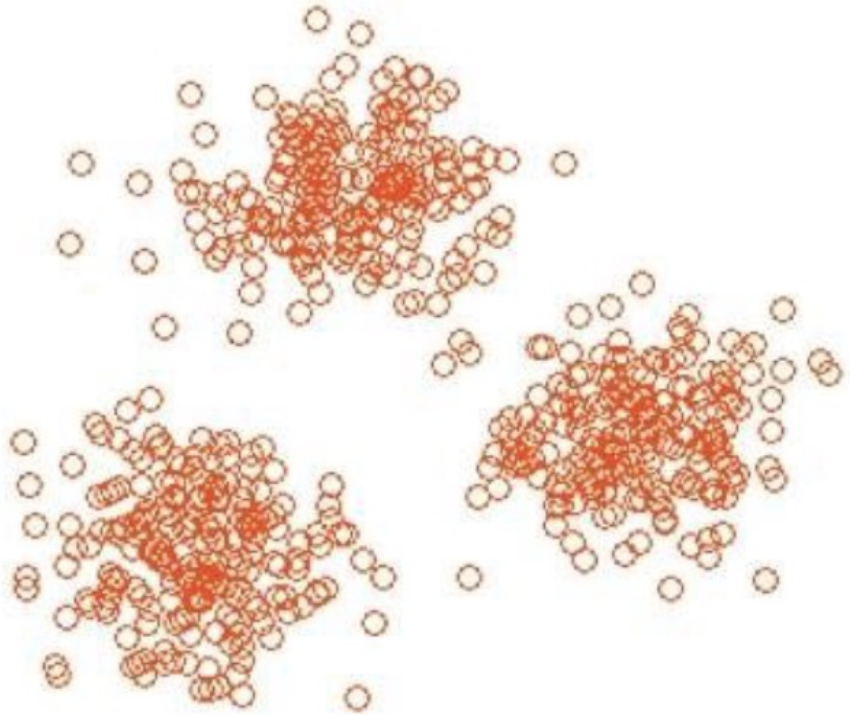
- Computational Economics:
  - predict if a stock will rise or fall
  - predict if a user will click on an ad or not
    - in order to decide which ad to show

# Unsupervised Learning

- Given data x, ask a good question ... about x or about model for x
- **Clustering**
  Find a set of prototypes representing the data
- **Principal Components**
  Find a subspace representing the data
- **Sequence Analysis**
  Find a latent causal sequence for observations
  - Sequence Segmentation
  - Hidden Markov Model (discrete state)
  - Kalman Filter (continuous state)
- **Hierarchical representations**
- **Independent components / dictionary learning**
  Find (small) set of factors for observation
- **Novelty detection**
  Find the odd one out

slide by Alex Smola

# Clustering

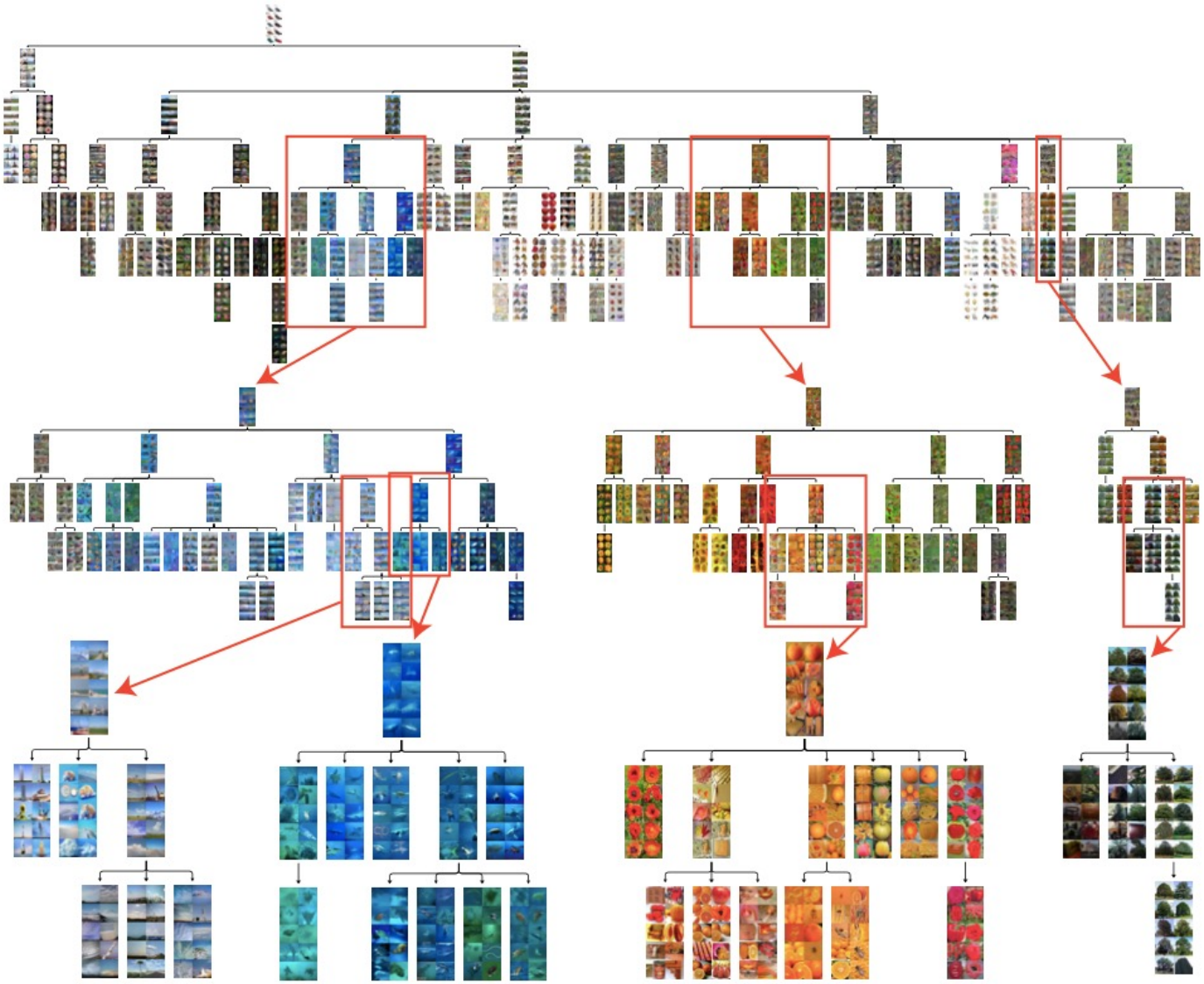


- Documents
- Users
- Webpages
- Diseases
- Pictures
- Vehicles

...

slide by Alex Smola

# Hierarchical Grouping



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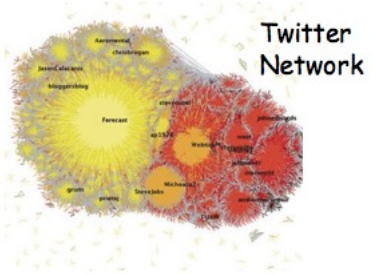
# Other Machine Learning Paradigm

## Clustering: discovering structure in data (only unlabeled data)

- E.g, cluster users of social networks by interest (community detection).



Facebook network



Twitter Network

## Semi-Supervised Learning: learning with labeled & unlabeled data

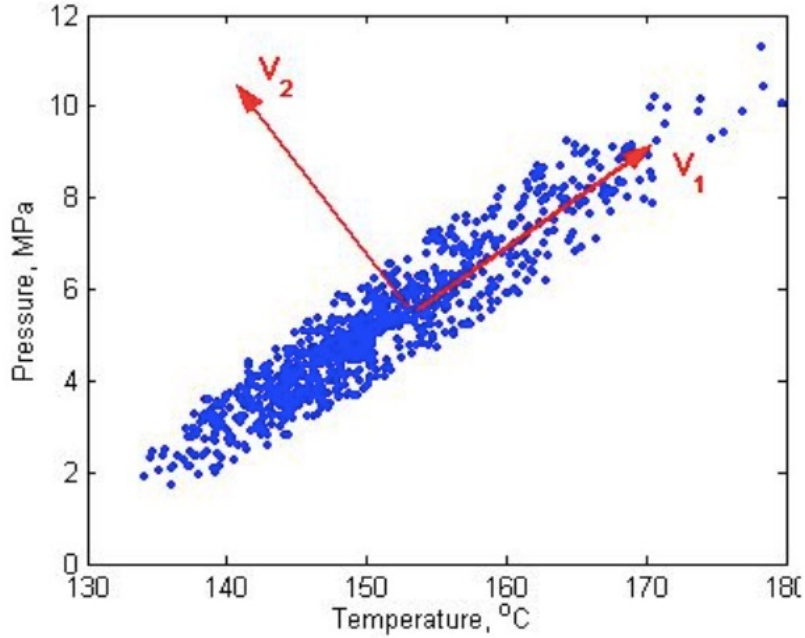
## Active Learning: learns pick informative examples to be labeled

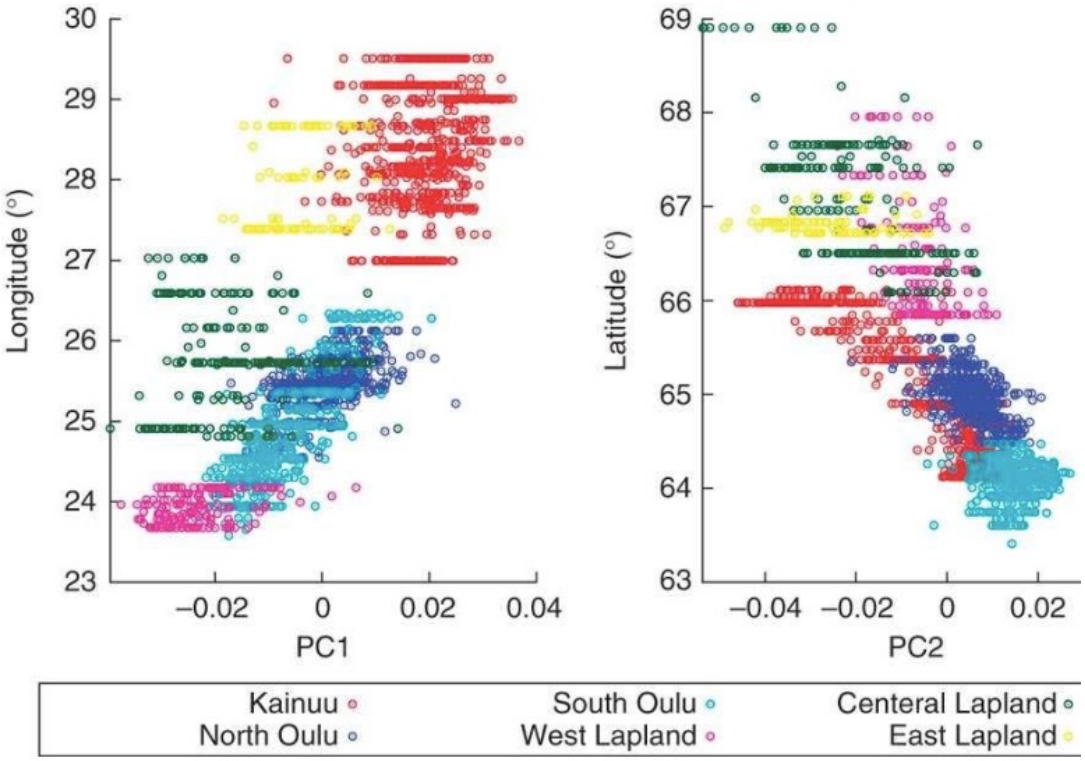
## Reinforcement Learning (acommodates indirect or delayed feedback)

## Dimensionality Reduction

## Collaborative Filtering (Matrix Completion), …

# Principal Components




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[Variance component model to account for sample structure in genome-wide association studies](), **Nature Genetics 2010**

# Important challenges in ML

- How important is the actual learning algorithm and its tuning
- Simple versus complex algorithm
- Overfitting
- Model Selection
- Regularization